QATARENERGY — RAS LAFFAN LNG COMPLEX

LIVE PLANT / EXPANSION INTERFACE RISK ASSESSMENT

CONTROLLED DOCUMENT | ACTIVITY-BASED HAZARD AND RISK REGISTER | PUBLIC-SOURCE PRE-ASSESSMENT

DOCUMENT STATUS: NOT APPROVED FOR EXECUTION — SITE DATA, WORK PACK AND OWNER REQUIREMENTS MUST BE VERIFIED

Public-source pre-assessment — site validation and authorized approval required. This document follows a generic professional reference format for layout guidance; it is NOT an official QatarEnergy/QatarEnergy LNG form, is NOT QatarEnergy-approved, and is not evidence of a site inspection.

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1. DOCUMENT CONTROL & PROJECT INFORMATION

Complete every applicable field with verified information. Use "Not applicable" only where justified; do not invent missing site information.

Project / Facility QatarEnergy Ras Laffan LNG / gas-processing complex — LNG trains area, Ras Laffan Industrial City (RLIC), State of Qatar	**Client / Asset Owner** QatarEnergy (State of Qatar energy company); operating company: QatarEnergy LNG [Q-03]. QatarEnergy is also owner, regulator and operator of Ras Laffan Port [Q-02]	**Principal Contractor** [SITE INPUT REQUIRED — EPC/main contractor(s) for the selected package are package-specific; NFS onshore EPC was awarded to a Technip Energies / CCC joint venture in 2023 [Q-05]; confirm the contractor for THIS package]	**Subcontractor(s)** [SITE INPUT REQUIRED — not publicly disclosed]
Assessment Title / Activity Live plant / expansion interface risk assessment — operations support, maintenance, construction, tie-in, pre-commissioning and commissioning activities interfacing with operating LNG trains, where confirmed in the approved work pack	**Work Location / Area** Ras Laffan LNG complex, RLIC [CONFIRM exact train / unit / plot with the approved plot plan]	**RA Number / Document ID** QELNG-RL-RA-001 [TO BE CONFIRMED BY DOCUMENT CONTROL]	**Revision / Status** Rev. 0 — Public-source pre-assessment — NOT APPROVED FOR EXECUTION
Assessment Date 28 October 2026 (public-source research snapshot)	**Review / Expiry Date** [PER PROJECT PROCEDURE — to be set at approval; review triggers are listed in Section 15]	**Related PTW / JSA / Method Statement** [SITE INPUT REQUIRED — verified references to be entered from the site PTW register and the approved work pack]	**Applicable Procedure / Standard** Owner HSE / PTW procedures [NOT REVIEWED — controlled documents]; ISO 45001:2018 + Amd 1:2024 (framework) [Q-07]; Qatar Ministerial Decision No. (17) of 2021 — heat stress [Q-06]
Prepared By / Role [REAL NAME / TITLE / SIGNATURE — to be completed by an authorized competent person]	**Reviewed By / Role** [REAL NAMES / TITLES — Operations, Construction, Process/Mechanical/Electrical, Marine/Port, HSE]	**Approver / Role** [AUTHORIZED APPROVER ONLY — this Rev. 0 pre-assessment is NOT approved]	**Workforce / Shifts** Current site headcount NOT PUBLICLY DISCLOSED — obtain gate / access-control and contractor mobilisation records (Section 5)

1.1 Document control detail

Document control item	Entry
Document number	QELNG-RL-RA-001 [TO BE CONFIRMED BY DOCUMENT CONTROL]
Project / contract number	[CONFIRM — to be assigned from the actual work package]
Facility / area	Ras Laffan LNG complex (LNG trains area), Ras Laffan Industrial City, Qatar — exact plot/area [CONFIRM WITH APPROVED PLOT PLAN]
Assessment boundary	The QatarEnergy LNG liquefaction / gas-processing complex at Ras Laffan only: operating LNG trains, associated gas treatment, condensate/LPG/NGL handling within the LNG complex, LNG storage and the Ras Laffan Port LNG berth interface, plus North Field East (NFE) and North Field South (NFS)

Document control item	Entry
	expansion interfaces inside that complex. EXCLUDES (separate facilities in RLIC — neighbours, not in-scope plant): Laffan Refineries (condensate refining), Pearl GTL and other gas-to-liquids plants, petrochemical plants, Barzan Gas, helium plants as standalone facilities, Mesaieed Industrial City, and offshore North Field production facilities. Interfaces with excluded neighbours are managed only as SIMOPS neighbours where the site plan shows one.
Work package / work order	[CONFIRM — assessment applies only to activities confirmed in the approved work pack]
Revision	Rev. 0 — Public-source pre-assessment
Assessment (issue) date	28 October 2026 (public-source research snapshot)
Date of site walkdown / work-pack review	[TO BE COMPLETED — no site visit has been made for this pre-assessment]
Prepared by	[REAL NAME / TITLE / SIGNATURE — to be completed by an authorized competent person]
Reviewed by	[REAL NAMES / TITLES / SIGNATURES — Operations, Construction, Process/Mechanical/Electrical, Marine/Port, HSE]
Approved by	[REAL NAME / TITLE / SIGNATURE — authorized approver only. This document is NOT approved.]
Classification	Working pre-assessment — not an official QatarEnergy document; not QatarEnergy-approved; contains no QatarEnergy branding and claims no review of controlled QatarEnergy/QatarEnergy LNG manuals.

VALIDATION DISCLAIMER: This assessment was prepared from publicly available sources only (see Section 17 references, access date 28 October 2026). No site visit, worker interview, permit review, drawing review or headcount verification has taken place. Every site-specific value that could not be verified from public sources is marked [CONFIRM] / [SITE INPUT REQUIRED] and must be established from controlled site documents and a competent site walkdown before this document is approved or used to authorize work.

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2. Executive summary

Why this assessment exists. QatarEnergy LNG operates 14 LNG trains at Ras Laffan Industrial City with a total production capacity of 77 MTPA [Q-03][Q-04], while the North Field East (NFE, 4 new mega trains, +32 MTPA) and North Field South (NFS, 2 new mega trains, +16 MTPA) expansion projects are under construction at the same complex [Q-03][Q-05]; QatarEnergy confirms the LNG expansion at Ras Laffan is under execution [Q-01]. Construction, tie-in, pre-commissioning and commissioning activity therefore interfaces with a live cryogenic hydrocarbon plant. That operating/expansion interface is the highest-risk feature of the complex and is the focus of this assessment.

Top major-accident hazards identified (Section 7): loss of containment of flammable gas/condensate with jet fire, flash fire or vapour cloud explosion (RA-03); LNG/refrigerant release from liquefaction systems, whose inventories and isolation details are NOT public and are therefore verification-critical (RA-04); cryogenic exposure at approx. -162 °C LNG conditions (RA-05, [Q-04]); possible toxic constituents in process streams, unquantified in public sources (RA-06); line opening on incompletely isolated systems (RA-09); heavy lifting for expansion interfaces near live plant (RA-13); first hydrocarbon introduction and start-up of new trains (RA-18); and breakdown of the emergency-response interface for a large transient workforce (RA-24). Extreme heat stress is a critical, legally regulated hazard under Qatar law (RA-21, [Q-06]).

Operating-status caution. Secondary reporting in 2026 described a production halt and phased restart at Ras Laffan following attacks on facilities, with damage reported to two LNG trains [Q-09]. Those reports are NOT verified against an official QatarEnergy statement in this pre-assessment. The current status of each train — live, shut down, under repair or under commissioning — materially changes the applicable hazards and is a Critical site input (Section 14). No control decision in this document relies on the press reports.

Critical controls and hold points. Work must not start until, as a minimum: the SIMOPS interface is reviewed and authorised by operations; isolations are proven with zero-energy verification; refrigerant inventories and stream composition data have been received and the RA-04/RA-06 decisions recorded; the owner risk matrix has replaced the provisional matrix in Section 6; the approved emergency plan details are briefed to every worker; and actual shift headcount is being captured for muster reconciliation. Each hold point is stated on the relevant register row and in Section 8.

Unresolved inputs. The current total headcount inside the complex boundary is NOT publicly disclosed and is not estimated anywhere in this document. Process conditions, refrigerant inventories, hazardous-area classification, exact workfronts and the current work package are likewise unconfirmed. Section 14 lists every site input required before competent review and approval. Residual risk is therefore NOT declared acceptable by this pre-assessment: residual ratings are provisional professional judgements that only become valid when the stated controls are implemented, verified and accepted by the authorized owner representative.

3. Project profile and source-backed facts

Labels used throughout: 'Verified public fact' = checked against the cited source on 28 October 2026; 'Target' = a published plan, not a current measured value; 'Technical consideration — not yet confirmed at site' = a credible hazard or control topic that site data must confirm or rule out; 'Site input required' = information that only controlled site documents can supply.

Fact	Value / statement	Status & source	What it does — and does not — prove
Operator entities	QatarEnergy (State of Qatar energy company) is the asset owner; QatarEnergy LNG (established 1984) operates the LNG ventures. QatarEnergy is also owner, regulator and operator of Ras Laffan Port.	Verified public fact [Q-02][Q-03]	Establishes the real operator entities. Does NOT establish day-to-day permit authority for a specific package — site organisation required.
Location	Ras Laffan Industrial City (RLIC), northeastern Qatar, approx. one hour by road north of Doha; RLIC area 295 km2; the city is the onshore base for processing North Field offshore gas. Ras Laffan Port lies 67 km from the North Field.	Verified public fact [Q-01][Q-02]	Establishes location and context. Does NOT establish the assessment's exact plot boundary — plot plan required.
Current LNG capacity / trains	14 LNG trains operating with total capacity 77 MTPA; six are mega-trains at 7.8 MTPA each. Train groups: N(1) 3x3.3 MTPA; N(2) 2x7.8; N(3) 1x7.8; N(4) 1x7.8; S(1) 2x3.3; S(2) 3x4.7; S(3) 2x8 MTPA.	Verified public fact [Q-03][Q-04]	Proves a large live liquefaction complex. Train-by-train CURRENT operating status (incl. any 2026 repair scope) is not stated on those pages — see the status caution in Section 2 and [Q-09].
LNG process (published description)	Condensate separation/stabilisation; staged removal of sulphur compounds, CO2 and water; chilling with propane and mixed refrigeration processes; main cryogenic heat exchanger to approx. -150 °C; pressure let-down to -162 °C; nitrogen removal; storage and vessel loading. Associated products: condensate,	Verified public fact [Q-04]	Grounds the cryogenic and refrigerant hazards (RA-04/RA-05) and confirms sulphur compounds are handled in gas treatment (RA-06 context). Publishes NO pressures, refrigerant inventories or stream compositions — none are assumed.

Fact	Value / statement	Status & source	What it does — and does not — prove
	LPG, ethane, sulphur, NGL. Al Khaleej Gas AKG-1/2 supply approx. 2 bscfd domestic sales gas.		
Expansion under construction — NFE	North Field East: 4 new LNG mega trains, +32 MTPA production entitlement; joint ventures established 2022, under construction.	Verified public fact [Q-03]	Defines the principal expansion interface inside the boundary. Does NOT prove which NFE workfronts are active today.
Expansion under construction — NFS	North Field South: 2 new LNG mega trains of 8 MTPA each (+16 MTPA); onshore EPC awarded 2023 to Technip Energies / CCC JV (approx. US$10bn); scope includes gas treatment, NGL recovery, helium extraction/refining, CCS facilities and a jetty boil-off gas recovery system, within RLIC.	Verified public fact — official announcement as reported [Q-05]; construction status [Q-03]	Defines the second expansion interface and the jetty boil-off scope relevant to RA-19. Package-level activity today must be confirmed from the work pack.
Capacity after NFE + NFS	77 MTPA rising to 126 MTPA when NFE and NFS are complete.	TARGET / capacity on completion [Q-03][Q-05]	A completion outcome, NOT today's capacity or output. Must not be used as current production data.
Further expansion — North Field West	Announced February 2024: a new LNG project from the North Field western sector of about 16 MTPA, taking Qatar's total to about 142 MTPA when completed before the end of 2030.	TARGET — announced plan [Q-08]	Expansion context only. Onshore scope, location detail and schedule at Ras Laffan are not defined in the sources opened; no hazard rows are built on NFW.
Marine / jetty interface	Ras Laffan Port (est. 1996; 56 km2) currently includes six LNG berths, six liquid product berths, one sulphur berth, six dry cargo berths, 14 support vessel berths and two offshore single point moorings; accommodates Q-Flex/Q-Max carriers. QatarEnergy LNG operates a jetty boil-off gas recovery facility and Ras Laffan Terminal Operations.	Verified public fact [Q-02][Q-03]	Justifies including the LNG berth / ship–shore interface (RA-19) inside the stated boundary. Berth-level work scope still requires Port Operations confirmation.
Neighbouring facilities — EXCLUDED	RLIC also hosts separate facilities outside this boundary, including the Laffan Refineries (condensate refining, 306,600 bpsd [Q-03]), Pearl GTL and other GTL plants, petrochemical plants, Barzan Gas and standalone helium facilities.	Verified public fact [Q-01][Q-03] (facility existence); exclusion is this assessment's boundary decision	Named so they are NOT silently merged into this RA. Their hazards, headcounts and emergency arrangements are not assessed here; only SIMOPS neighbour interfaces, if any, are managed through RA-02.
Heat regulation (Qatar)	Ministerial Decision No. (17) of 2021: outdoor work prohibited 10:00–15:30 in open areas and unventilated shaded areas, annually 1 June–15 September; work must be suspended where WBGT exceeds 32.1 °C; duties include displayed schedules, cold water, shaded rest, medical checks and trained first-aiders.	Verified public fact — QATAR LAW as reported by the Ministry of Labour [Q-06]	Sets the legal floor for RA-21. The current official text must be confirmed with the Ministry / site legal register.
Management-system framework	ISO 45001:2018, confirmed in 2024, remains current; Amendment 1:2024 (climate action) applies; a revision is in progress (stage 90.92 at access date).	Verified public fact [Q-07]	OH&S management framework only. NOT a hazard checklist and NOT Qatar law.
2026 disruption reporting	Reuters reported (April 2026) a March 2026 production halt at Ras Laffan after attacks on facilities and a planned phased restart; other reporting attributes to QatarEnergy statements of damage to two LNG trains totalling 12.8 MTPA.	SECONDARY REPORTING — NOT VERIFIED against an official statement [Q-09]	Raises a question, proves nothing for control purposes. Train-by-train status is a Critical site input (Section 14). No figure from these reports is used elsewhere in this document.

4. Site footprint, asset and workfront register

Public sources describe the facility elements below at complex level. No public source provides a current approved workfront register, package split or plot boundary; every operational field is therefore [SITE INPUT REQUIRED]. Excluded neighbours are listed only to record their exclusion.

Facility / asset / workfront	Location	Package / contractor	Current activity & SIMOPS interface	Operational status	Source / drawing reference	Data owner	Verification status
Operating LNG trains — N-series (N(1) to N(4), 7 trains)	Ras Laffan LNG complex [CONFIRM train/plot IDs]	[CONFIRM package]	Live operation; maintenance/construction interface where tie-ins or modifications occur	Operating per operator profile [Q-03]; current status of each train [CONFIRM — see Q-09 caution]	Operator profile [Q-03]; plot plan [SITE INPUT REQUIRED]	Operations	Capacity verified publicly; live status not verified
Operating LNG trains — S-series (S(1) to S(3), 7 trains)	Ras Laffan LNG complex [CONFIRM train/plot IDs]	[CONFIRM package]	Live operation; maintenance/construction interface as above	Operating per operator profile [Q-03]; current status of each train [CONFIRM]	Operator profile [Q-03]	Operations	Capacity verified publicly; live status not verified
Gas treatment, condensate	LNG complex	[CONFIRM]	Process operations supporting the trains;	Operating (process	Operator operations page [Q-	Operations	Process route verified;

Facility / asset / workfront	Location	Package / contractor	Current activity & SIMOPS interface	Operational status	Source / drawing reference	Data owner	Verification status
stabilisation, LPG/NGL handling within the LNG complex	[CONFIRM units]		intrusive-work interface	described publicly [Q-04])	04]		conditions NOT public
LNG storage and loading; jetty boil-off gas recovery	LNG complex / port interface [CONFIRM tank and berth IDs]	[CONFIRM]	Storage, loading and boil-off recovery operations; berth maintenance interface	Operating [Q-03][Q-04]	Operator pages [Q-02][Q-03] [Q-04]	Operations / Port Operations	Functions verified; tank/berth details not public
Ras Laffan Port — six LNG berths (ship–shore interface only)	Ras Laffan Port, RLIC [Q-02]	Port Operations	LNG carrier loading; jetty work interface (RA-19)	Operating [Q-02]	QatarEnergy Ports Information [Q-02]	Port Operations	Berth count verified; work scope to confirm
North Field East (NFE) — 4 new mega trains under construction	Within RLIC LNG expansion area [CONFIRM plots]	EPC package(s) [CONFIRM contractor for this RA's package]	Construction / pre-commissioning / commissioning adjoining live plant	Under construction [Q-03]	Operator profile [Q-03]	Construction Manager	Status verified publicly; workfronts not verified
North Field South (NFS) — 2 new mega trains under construction	Within RLIC [CONFIRM plots]	Onshore EPC: Technip Energies / CCC JV (2023 award) [Q-05] [CONFIRM current package split]	Construction incl. gas treatment, NGL recovery, helium, CCS and jetty boil-off facilities	Under construction [Q-03][Q-05]	QatarEnergy announcement as reported [Q-05]	Construction Manager	Scope verified publicly; workfronts not verified
Al Khaleej Gas (AKG-1/2) domestic sales gas trains	LNG complex [CONFIRM]	[CONFIRM]	Pipeline sales gas production (~2 bscfd) — interface only where the work pack touches it	Operating [Q-04]	Operator operations page [Q-04]	Operations	Included as an interface, not a separate assessment
Laffan Refineries; Pearl GTL / other GTL; petrochemical plants; Barzan Gas; standalone helium plants	RLIC — SEPARATE FACILITIES	n/a	NONE — excluded from this assessment; SIMOPS neighbour interface only if the site plan shows one	Separate operations [Q-01][Q-03]	Operator pages [Q-01][Q-03]	n/a	EXCLUDED — separate assessments by their operators
Offshore North Field production facilities; Mesaieed Industrial City	Offshore / Mesaieed — SEPARATE LOCATIONS	n/a	NONE — excluded from this assessment	Separate operations	QatarEnergy pages [Q-01][Q-02]	n/a	EXCLUDED

5. Workforce and exposure register

5.1 Public figures — with date, scope and correct use

Figure	Value	Date / scope	Correct use in this assessment
QatarEnergy LNG production capacity (context figure — NOT a workforce figure)	77 MTPA from 14 LNG trains	QatarEnergy LNG corporate profile, current at access date [Q-03][Q-04]	Capacity context ONLY. Included here to prevent misuse: capacity figures say nothing about how many people are on site.
Total persons physically inside the Ras Laffan LNG complex boundary on the issue date	NOT PUBLICLY DISCLOSED	Current — not found in any public source reviewed	MUST be obtained from the gate/access-control report and contractor mobilisation/daily manpower reports. Do not estimate. RLIC-wide population or accommodation figures, where published by third parties, are NOT a complex headcount and are not used.
Number of active workfronts / packages on the issue date	NOT CONFIRMED IN PUBLIC SOURCES	Current	Public sources confirm NFE and NFS construction is under way [Q-03][Q-05] and that the LNG expansion at Ras Laffan is under execution [Q-01]; the live workfront register must come from site management.
Expansion construction workforce	NOT PUBLICLY DISCLOSED in the sources opened	Current	Third-party project summaries publish workforce estimates for individual EPC packages; none was verified against an official QatarEnergy/QatarEnergy LNG source in this check, so no figure is used.

WARNING: LNG capacity (MTPA) is not a workforce figure, and no RLIC-wide population, accommodation or third-party EPC estimate may be used as the number of people inside the LNG complex boundary. Muster, welfare, transport and emergency planning require the actual gate/mobilisation records below.

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5.2 Actual manpower and exposure register — TO BE COMPLETED FROM SITE RECORDS

Source of count required: gate access-control / badge report, muster roster, contractor mobilisation register or daily manpower report. Until a dated record is attached, every field below remains 'Not publicly disclosed — obtain actual site record.'

Field	Value (from site record only)	Record reference
Headcount date/time of count; shift pattern in force (day/night) [CONFIRM]	Not publicly disclosed — obtain actual site record	
Direct operator (QatarEnergy LNG) personnel inside boundary — day / night	Not publicly disclosed — obtain actual site record	
EPC / main contractor personnel (NFE and NFS packages listed separately) — day / night	Not publicly disclosed — obtain actual site record	
Subcontractor personnel (each subcontractor listed separately) — day / night	Not publicly disclosed — obtain actual site record	
Vendor / OEM / commissioning personnel; visitors inside boundary	Not publicly disclosed — obtain actual site record	
Marine / jetty and port-interface personnel inside boundary	Not publicly disclosed — obtain actual site record	
TOTAL persons inside the defined boundary (this count); peak concurrent workforce in the period	Not publicly disclosed — obtain actual site record	
Work-at-height personnel; confined-space entrants and attendants (this shift)	Not publicly disclosed — obtain actual site record	
Lifting crew; hot-work crew and fire watches (this shift)	Not publicly disclosed — obtain actual site record	
Electrical / instrument crew; plant/traffic interface personnel (this shift)	Not publicly disclosed — obtain actual site record	
Emergency-response personnel on shift (fire, medical, rescue)	Not publicly disclosed — obtain actual site record	
Prepared by (name/role) / Verified by (name/role) / Revision & reporting period	Not publicly disclosed — obtain actual site record	

6. Assessment methodology and applicable requirements

Method. Work is decomposed into activity steps (Section 7). For each step the assessor identifies the hazard source and the specific hazardous event, the persons exposed and how, and the credible consequences — including fatality and major-accident consequences where they are foreseeable. Controls already in place are recorded separately from additional controls, and each existing control is marked with its evidence status: in this pre-assessment almost all are 'to be verified', because no site evidence has been reviewed. Additional controls follow the hierarchy of controls (elimination, substitution, engineering, administrative, PPE), prioritising isolation and engineering barriers over procedures and PPE. A risk rating is never reduced merely by writing 'follow procedure' or adding PPE.

Three different risk values. (1) Initial/inherent risk assumes no controls beyond those physically inherent in the task. (2) Current risk credits only existing controls that are implemented AND verified by evidence. (3) Residual risk is rated only after the additional controls are implemented and independently/competently checked. In this draft, initial and residual ratings are PROVISIONAL PROFESSIONAL JUDGEMENTS made for comparison and prioritisation; they must be re-rated in the site validation workshop.

Major-accident limitation. A 5x5 matrix is a screening and prioritisation tool. It is NOT sufficient on its own for major-accident process hazards: a low likelihood score must never be used to dismiss a catastrophic scenario (RA-03, RA-04, RA-06). Such scenarios require the project's specialist studies (HAZID, HAZOP, LOPA, QRA, bow-tie) and the interface to those studies is a site input (Section 14). This document is an activity-based HSE risk assessment; it does not replace a process hazard analysis, a method statement, or task-level JSAs (Section 9).

6.1 Proposed 5 x 5 risk matrix — PROPOSED WORKING MATRIX ONLY

Use the client/project-approved risk matrix whenever one exists. The 5 x 5 matrix below is a PROPOSED reference method only and must be replaced or formally accepted by the responsible organization before use. Risk score = Likelihood x Severity. The owner risk matrix was not available; likelihood/severity definitions are proposed qualitative definitions pending approval.

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PROPOSED 5 x 5 RISK MATRIX

S x L	1	2	3	4	5
5	5	10	15	20	25
4	4	8	12	16	20
3	3	6	9	12	15
2	2	4	6	8	10
1	1	2	3	4	5

RATING DEFINITIONS (TO BE APPROVED)

LIKELIHOOD SCALE

Level	Term	Definition / guide
1	Rare	Exceptional; highly unlikely in the defined conditions.
2	Unlikely	Not expected, but credible.
3	Possible	Could occur under some circumstances.
4	Likely	Expected to occur in many circumstances.
5	Almost certain	Expected to occur frequently / repeatedly.

SEVERITY SCALE

Level	Term	Definition / guide
1	Insignificant	No injury or negligible harm / damage.
2	Minor	First-aid case or minor, reversible damage.
3	Moderate	Medical-treatment / lost-time potential; significant reversible harm or damage.
4	Major	Single-fatality potential, life-changing injury or major damage.
5	Catastrophic	Multiple fatalities or catastrophic asset / environmental consequences.

PROPOSED SCORING BANDS (subject to project approval): 1–4 Low — manage by routine controls. 5–9 Moderate — specific controls and supervision. 10–16 High — additional controls, documented review and authorized acceptance required before work. 17–25 Extreme — work must not proceed until risk is reduced and acceptance is given at the level the owner's system requires. These bands are proposed working bands, not a universal ISO/QatarEnergy matrix. High/Extreme residual risks in Section 7 carry explicit hold points.

6.2 Applicable requirements and verification register

Requirement / source	Jurisdiction & type	Relevant topic (summarised, not reproduced)	Applicability	Version/date checked	Owner/client document still needed	Verification status
Qatar Ministerial Decision No. (17) of 2021 — summer working hours / heat stress [Q-06]	Qatar — LAW / REGULATORY	No outdoor work 10:00–15:30 in open areas and unventilated shaded areas, annually 1 Jun–15 Sep; WBGT monitoring with suspension above 32.1 °C; displayed schedules, cold water, shaded rest, medical checks, trained first-aiders	All outdoor work in season	Decision of 2021; 2026 campaign reporting checked 28 October 2026	Current official text via Ministry of Labour / site legal register; site heat-stress procedure	Requirement verified via Ministry reporting; official text and site procedure outstanding
Qatar labour / OSH / environmental framework (general)	Qatar — LAW	Employer duties for worker safety, hours, welfare and environmental protection — exact current texts and permit conditions to be confirmed by the project's legal/HSE function	All work	Not verified in this check	Project legal register; site environmental permit	SITE INPUT REQUIRED — do not cite clause numbers from memory
ISO 45001:2018 + Amd 1:2024 [Q-07]	International — MANAGEMENT SYSTEM STANDARD	Framework for hazard identification, worker participation, operational control, emergency preparedness and review	Assessment process framework	Confirmed current 2024; checked 28 October 2026	Project HSE management plan mapping	Verified as framework only — not a legal requirement
Owner (QatarEnergy / QatarEnergy LNG) HSE and PTW requirements	Client-specific — CONTRACTUAL	PTW, isolation, lifting, confined space, gas testing, jetty/port, emergency and competency requirements	All work — governing documents for execution	Controlled documents; NOT reviewed	Current authorized copies via the project's document control	NOT REVIEWED — critical site input
OSHA PSM overview & hazard recognition [Q-10][Q-11]	United States — TECHNICAL GUIDANCE ONLY	Process-safety management concepts for hydrocarbon interface work	Technical reference for RA-01/03/04/06/07/18	As cited in the reference master prompt	Owner process-safety standards (governing)	Guidance only — NOT Qatar law
OSHA confined spaces; welding/cutting guidance [Q-12] [Q-13]	United States — TECHNICAL GUIDANCE ONLY	Control concepts for confined-space entry, hot work and welding fume	Technical reference for RA-10/11	As cited in the reference master prompt	Owner confined-space and hot-work procedures (governing)	Guidance only — NOT Qatar law

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7. Detailed activity-based risk assessment register

Use one entry for each meaningful work step / hazard scenario. Hazards are described specifically; existing controls are distinguished from additional controls; residual risk is rated only for controls that are implemented and verified. 25 specific entries (half-size pack). Ratings use the PROVISIONAL matrix in Section 6.1 and are professional judgements for prioritisation only — they must be re-rated in the site validation workshop with the owner matrix. 'Existing controls' in this pre-assessment are stated with their evidence status; a control without evidence is NOT credited in any residual rating until verified on site.

RA ID	Work activity / step	Asset / area / workfront	Hazard / hazardous event	Potential consequence & persons at risk	Existing controls	Initial risk (L x S = score)	Additional controls / actions (hierarchy of controls)	Responsible role & target date	Residual risk (L x S = score)	PTW / procedure reference	Verification evidence / status
RA-01	Construction / maintenance adjacent to operating LNG trains and gas-processing equipment	Live train areas [CONFIRM unit/plot]	Impact, dropped load, vehicle strike or hot particles contacting live hydrocarbon equipment, causing loss of containment (LOC).	Release, fire/explosion, multiple fatalities, major plant damage, escalation. Persons at risk: Construction crews, operators, adjacent contractors near live equipment.	TO BE VERIFIED: barricading, PTW and operations oversight are owner systems — no site evidence reviewed.	3 x 5 = 15 — HIGH	ELIMINATION: use planned shutdowns for interface work where feasible. ENGINEERING: physical barriers/impact protection; equipment-protection zones on the plot plan. ADMIN: SIMOPS review + written operations authorisation before work; daily coordination with Area Authority. PPE: task PPE per site standard (last line only).	Area Authority / Operations Rep. Priority: Critical Target date: [SITE INPUT REQUIRED]	2 x 5 = 10 — HIGH	Owner PTW; SIMOPS authorisation; plot plan extract Review trigger / reference: Review on plant-status/workfront change. OSHA PSM overview (guidance) [Q-10].	HOLD: no work inside the protection zone of live equipment without signed operations authorisation. Stop on any plant alarm/upset until cleared. [Evidence status: To be verified — no site evidence reviewed]
RA-02	SIMOPS: hot work / testing concurrent with hydrocarbon operations; multiple permits in one area	NFE/NFS interface areas adjoining live plant [CONFIRM]	Conflicting permits or incompatible simultaneous activities (hot work near venting, draining, line opening) creating ignition/exposure pathways no single permit holder can see.	Fire/explosion, toxic exposure, multiple fatalities; conflicting emergency actions. Persons at risk: All parties in the shared area — shared atmosphere, escape routes, permits.	TO BE VERIFIED: owner SIMOPS procedure and live permit register not reviewed.	3 x 5 = 15 — HIGH	ENGINEERING: single permit-coordination point with live permit map/board. ADMIN: SIMOPS matrix of prohibited combinations approved by operations; one accountable Area Authority per area/shift; conflict check at issue and each handover; suspension rule when operations activities change.	Area Authority / Operations Rep. Priority: Critical Target date: [SITE INPUT REQUIRED]	2 x 5 = 10 — HIGH	Owner PTW system; SIMOPS plan; permit register Review trigger / reference: Review each shift change and on any new permit in the area.	HOLD: hot work prohibited during hydrocarbon venting, draining, sampling or line opening in the same area unless a specific SIMOPS assessment signed by operations authorises it. [Evidence status: To be verified — no site evidence reviewed]
RA-03	Work on/near pressurised hydrocarbon gas and condensate systems (tie-ins, testing, reinstatement)	Trains, gas treatment, condensate systems [CONFIRM]	Uncontrolled flammable gas/condensate release forming a flammable cloud; jet fire, flash fire or vapour cloud explosion with escalation.	Multiple fatalities, major asset loss, environmental release — major-accident event. Persons at risk: Crews at the release point, downwind personnel, operators — thermal radiation, blast, projectiles.	TO BE VERIFIED: gas detection, ESD and fire-protection status at the interface not evidenced. A 5x5 matrix alone is insufficient here; HAZID/HAZOP/bow-tie required.	3 x 5 = 15 — HIGH	ELIMINATION: defer hydrocarbon introduction until construction in that system completes. ENGINEERING: positive isolation / double block and bleed per owner procedure; verified detection coverage; bonding/earthing. ADMIN: line-break/reinstatement procedure with independent verification; leak test before hydrocarbon introduction; emergency response on standby for first fill. PPE: FR clothing, personal gas monitor where the standard requires.	Process Engineer / Area Authority Priority: Critical Target date: [SITE INPUT REQUIRED]	2 x 5 = 10 — HIGH	Line-opening permit; isolation certificate; leak-test/PSSR records Review trigger / reference: Review on boundary/composition change. OSHA PSM hazard recognition (guidance) [Q-11].	HOLD: no hydrocarbon introduction without completed leak test, PSSR sign-off and operations acceptance. Residual HIGH — acceptance only by the authorised owner representative. [Evidence status: To be verified — no site evidence reviewed]
RA-04	LNG / refrigerant release during liquefaction-train work — VERIFICATION-CRITICAL	Trains — propane and mixed-refrigerant circuits [CONFIRM systems]	Release of LNG, propane or mixed refrigerant (operator confirms propane and mixed refrigeration processes are used [Q-04]; inventories, compositions and pressures are NOT public). Drifting flammable cloud; LNG pool formation and rapid phase transition on spill to water/concrete.	Flash fire / vapour cloud explosion, multiple fatalities, cryogenic structural damage, escalation to adjacent trains. Persons at risk: Train and maintenance crews, downwind workfronts — cloud drift, thermal radiation, embrittlement failures.	NOT CONFIRMED: refrigerant inventory, isolation segmentation and detection coverage are not public — obtain process safety information first.	3 x 5 = 15 — HIGH	VERIFY FIRST: refrigerant P&IDs, inventories, isolation philosophy, detection coverage plots. ENGINEERING: sectional isolation limiting releasable inventory; detection/ESD proven before intrusive work; drainage/containment routes confirmed clear. ADMIN: refrigerant-specific emergency actions briefed (cloud drift, muster upwind); no hot work in refrigerant areas without gas-free proof. PPE: FR + cryogenic PPE for breaking containment (RA-05).	Process Safety Engineer / Area Authority Priority: Critical Target date: [SITE INPUT REQUIRED]	2 x 5 = 10 — HIGH	Owner PTW; isolation certificate; refrigerant work procedure [SITE INPUT REQUIRED] Review trigger / reference: Review when process safety information is issued. Process description [Q-04].	HOLD: no intrusive work on refrigerant/LNG systems until inventories, isolation points and detection coverage are received and briefed. Residual HIGH. [Evidence status: NOT CONFIRMED — verification-critical]

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RA ID	Work activity / step	Asset / area / workfront	Hazard / hazardous event	Potential consequence & persons at risk	Existing controls	Initial risk (L x S = score)	Additional controls / actions (hierarchy of controls)	Responsible role & target date	Residual risk (L x S = score)	PTW / procedure reference	Verification evidence / status
RA-05	Breaking into or working near LNG and cryogenic refrigerant lines/equipment — cryogenic exposure	LNG storage/loading lines, cold train sections [CONFIRM]	Contact with LNG/refrigerant at cryogenic temperature (approx. -162 °C at atmospheric pressure per the operator's process description [Q-04]): cold burns, eye injury, brittle fracture of unsuitable materials, oxygen condensation effects near cold leaks.	Severe cold burns, permanent eye injury, embrittlement failure, secondary release. Persons at risk: Fitters, operators, inspectors at cold equipment — jet/pool contact, cold surfaces.	TO BE VERIFIED: cryogenic PPE issue, cold-work procedures and material specifications not reviewed.	3 x 4 = 12 — HIGH	ENGINEERING: insulation/guarding of cold surfaces in work areas; cryogenic-rated tools/materials only. ADMIN: depressurisation and warm-up proven before breaking containment (RA-09); non-essential personnel excluded during line-up changes; buddy system in cold areas. PPE: cryogenic gloves, face shield/eye protection, protective clothing per site standard for any LNG/refrigerant contact task.	Mechanical Supervisor / HSE Officer Priority: High Target date: [SITE INPUT REQUIRED]	2 x 4 = 8 — MODERATE	Owner PTW; line-opening permit where containment is broken Review trigger / reference: Review for every cold-system task. Operator process description [Q-04].	STOP: any visible cold leak, frost anomaly or unexplained vapour cloud stops work in the area until operations assess it. [Evidence status: To be verified — no site evidence reviewed]
RA-06	Work where toxic constituents may be present in process streams (sulphur compounds / H2S) — VERIFICATION-CRITICAL	Gas treatment areas — composition NOT public [CONFIRM from process data]	The operator confirms sulphur compounds are removed from the gas in stages [Q-04] but publishes no H2S concentration for any workfront. Toxic-gas exposure is credible but unquantified.	Acute toxic exposure, single/multiple fatalities, failed rescue by unprotected responders. Persons at risk: Crews opening or working near process systems — inhalation at the work point and in low areas.	NOT CONFIRMED: no composition, P&ID or SDS data available. Treat as unassessed until process safety information is supplied.	3 x 5 = 15 — HIGH	VERIFY FIRST: full stream composition (incl. H2S/mercaptans), hazardous-area drawings, owner gas-testing criteria — before intrusive work. ENGINEERING (if sour service confirmed): fixed detection, ventilation. ADMIN: toxic-gas training; personal monitors with alarm limits set ONLY to the approved site standard (none assumed here); buddy system; SCBA-equipped rescue plan. PPE: escape respirators/SCBA per site standard.	Process Safety / HSE Manager Priority: Critical Target date: [SITE INPUT REQUIRED]	2 x 5 = 10 — HIGH	Owner gas-testing procedure; toxic-gas contingency plan [SITE INPUT REQUIRED] Review trigger / reference: Review when composition data is issued. Process description [Q-04].	HOLD: intrusive work on any process system is prohibited until composition data is received and the toxic-gas controls decision is recorded by a competent person. [Evidence status: NOT CONFIRMED — verification-critical]
RA-07	Temporary bypass, override or impairment of safety-critical systems (detection, ESD, fire systems)	Any safety-critical system in the interface area [CONFIRM]	A credited protection barrier left bypassed/impaired without management of change during high-risk work.	Undetected release escalation, major fire/explosion, multiple fatalities. Persons at risk: All personnel in the affected area — loss of the detection/protection barrier.	TO BE VERIFIED: no bypass/impairment register reviewed.	2 x 5 = 10 — HIGH	ADMIN: bypasses only under formal MOC/impairment permit — named owner, time limit, compensating measures (manned monitoring, portable detection), restoration verification; register reviewed each handover; overdue items escalated to plant management. ENGINEERING: restore and function-test before associated hot/intrusive work resumes.	Process Safety Engineer Priority: Critical Target date: [SITE INPUT REQUIRED]	1 x 5 = 5 — MODERATE	MOC / impairment permit; bypass register Review trigger / reference: Review every shift while any bypass is active. OSHA PSM overview (guidance) [Q-10].	HOLD: hot work and line opening prohibited where the credited barrier is bypassed without approved compensating measures in place. [Evidence status: To be verified — no site evidence reviewed]
RA-08	Isolation and lockout-tagout (LOTO) of mechanical / electrical / instrument energy	All systems in package [CONFIRM isolation points from P&IDs]	Failure to identify and isolate all energy sources (pressure, electrical, stored, gravity, chemical) — unexpected start-up or release during work.	Fatality/serious injury from pressure release, moving equipment, shock or falling load. Persons at risk: Fitters, electricians, instrument technicians at the work point.	TO BE VERIFIED: owner LOTO/isolation procedure and isolation hardware availability not evidenced.	3 x 5 = 15 — HIGH	ENGINEERING: positive isolation (disconnection/blinding) for line breaking per owner procedure; lockable isolation points. ADMIN: isolation certificate listing every energy source, applied by an authorised isolator; personal locks/tags per worker; zero-energy verification (try-start, test-for-dead, bleed check) recorded before work; group lockout for multi-craft work.	Authorised Isolator / Area Authority Priority: Critical Target date: [SITE INPUT REQUIRED]	1 x 5 = 5 — MODERATE	Isolation certificate; LOTO register; PTW Review trigger / reference: Review on any isolation-boundary change. Owner LOTO procedure [SITE INPUT REQUIRED].	HOLD: work must not start until the zero-energy state is proven and recorded; any doubt about an energy source stops the job. [Evidence status: To be verified — no site evidence reviewed]
RA-09	Line opening / flange breaking on process and utility piping	Tie-in and modification points [CONFIRM from line list]	Residual pressure, trapped gas/liquid or wrong-line identification releasing hydrocarbons, refrigerant or chemicals at the	Fire, toxic/cryogenic exposure, fatality, environmental release. Persons at risk: Pipefitters at the flange face; personnel in the line of fire.	TO BE VERIFIED: line-opening procedure, drain/vent provisions and lineup drawings not reviewed.	4 x 5 = 20 — EXTREME	ELIMINATION: break only at designated break points in the approved procedure. ENGINEERING: drain, vent, purge and prove depressurised and temperature-safe at the break point; first-break technique from a protected position (far side first, clear of the line of fire). ADMIN: line ID independently verified against line	Mechanical Supervisor / Area Authority Priority: Critical Target date: [SITE INPUT REQUIRED]	2 x 5 = 10 — HIGH	Line-opening permit; isolation certificate; gas test certificate Review trigger / reference: Review for every new line	HOLD: first break prohibited until drain/vent proven, gas test satisfactory and permit signed. Stop on any unexpected pressure/product at the break.

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			break point; line-of-fire ejection.				list/P&ID by two competent persons; gas test at the break point; permit signed by operations. PPE: face/cryogenic/chemical protection per fluid data.			break. Owner line-opening procedure [SITE INPUT REQUIRED].	[Evidence status: To be verified — no site evidence reviewed]
RA-10	Hot work (welding, cutting, grinding) in or near hydrocarbon plant — incl. fume and arc exposure	Fabrication areas and in-situ hot work locations [CONFIRM]	Sparks/slag igniting flammable atmosphere, combustibles or hydrocarbon residues; fire spreading beyond the work point. Welding fume, UV/IR radiation and burns at the work point.	Fire/explosion, fatalities, major plant damage; welder/bystander eye injury, fume exposure. Persons at risk: Hot-work crew, fire watch, nearby workers and plant; helpers exposed to arc and fume.	TO BE VERIFIED: hot-work procedure, gas-testing criteria, fire-watch practice and fume controls not reviewed.	3 x 5 = 15 — HIGH	ELIMINATION: prefabricate in a designated safe fabrication area wherever possible. ENGINEERING: spark containment (blankets/screens); combustibles removed/isolated within the radius set by the site procedure; local exhaust ventilation for enclosed welding. ADMIN: hot-work permit with pre-work and continuous gas testing where required; trained fire watch during work and for the post-work fire-check period in the site procedure; extinguishers verified at the point of work; fume assessment from consumable SDS. PPE: welding PPE; respiratory protection per assessment.	Hot Work Supervisor / Fire Watch Priority: Critical Target date: [SITE INPUT REQUIRED]	1 x 5 = 5 — MODERATE	Hot-work permit; gas test certificate; fire-watch record Review trigger / reference: Review for every hot-work permit. OSHA welding guidance [Q-12]; owner procedure [SITE INPUT REQUIRED].	HOLD: hot work stops immediately on gas-test alarm, loss of fire watch, or any plant upset alarm in the area. [Evidence status: To be verified — no site evidence reviewed]
RA-11	Confined-space entry into vessels, columns, tanks, sumps or large-bore equipment	Equipment in package requiring internal work [CONFIRM entry list]	Toxic/flammable/ oxygen-deficient atmosphere, engulfment or heat stress inside the space; failed rescue multiplying casualties.	Single or multiple fatalities (entrant and would-be rescuers). Persons at risk: Entrants, attendant, rescue team — atmosphere inside the space.	TO BE VERIFIED: confined-space procedure, equipment classification and rescue capability not evidenced.	3 x 5 = 15 — HIGH	ELIMINATION: remote inspection/cleaning where practicable. ENGINEERING: isolation and blinding of all connected lines; cleaning/purging proven by test; continuous ventilation and atmospheric monitoring. ADMIN: entry permit with named attendant, communications check, entry log, rehearsed rescue plan with equipment at the entry point before entry starts; training verified. PPE: respiratory protection per test results and site procedure.	Confined-Space Entry Supervisor Priority: Critical Target date: [SITE INPUT REQUIRED]	1 x 5 = 5 — MODERATE	Confined-space entry permit; gas test certificate; rescue plan Review trigger / reference: Review for every entry. OSHA Confined Spaces (guidance) [Q-13]; owner procedure [SITE INPUT REQUIRED].	HOLD: no entry without rescue capability physically in place; entry stops on monitor alarm, ventilation failure or attendant leaving post. [Evidence status: To be verified — no site evidence reviewed]
RA-12	Electrical work, temporary power distribution and backfeed risks	Substations, MCCs, field installations, construction areas [CONFIRM voltages from drawings]	Contact with live parts, arc flash, backfeed from temporary generators/UPS/interconnections; damaged trailing cables in dusty/wet conditions causing shock or fire. Voltages are NOT stated here — obtain from approved drawings.	Electrocution, arc-flash burns, fatality; electrical fire. Persons at risk: Electricians, instrument technicians, all users of portable tools.	TO BE VERIFIED: electrical safety rules, authorised-person register, drawings and temporary-power inspection regime not reviewed.	3 x 5 = 15 — HIGH	ENGINEERING: isolation at all sources incl. temporary supplies; earthing where the rules require; RCD/earth-leakage protection and protected cable runs for temporary power. ADMIN: work only by electrically authorised persons; prove dead with a verified tester before touching; electrical permit for HV work as defined by site rules; temporary-supply register; alterations only by authorised electricians; defective items removed immediately. PPE: arc-rated clothing, voltage-rated gloves per site standard.	Authorised Electrical Person / E&I Engineer Priority: Critical Target date: [SITE INPUT REQUIRED]	1 x 5 = 5 — MODERATE	Electrical permit / sanction for test; isolation certificate; temporary installation certificates Review trigger / reference: Review on any supply-configuration change. Owner electrical rules [SITE INPUT REQUIRED].	HOLD: no contact work until prove-dead is recorded; any unidentified cable is treated as live and stops the job. [Evidence status: To be verified — no site evidence reviewed]
RA-13	Heavy and critical crane lifts — equipment, modules and spools for expansion interfaces (incl. crane set-up on prepared ground)	Laydown areas and plant installation points [CONFIRM]	Rigging failure, wrong load weight/centre of gravity, ground-bearing failure (outrigger sinking into uncompacted ground or over buried services), or crane malfunction — dropped/swinging	Fatality, serious injury, impact damage to live plant with secondary hydrocarbon release. Persons at risk: Rigging crew and personnel in the drop/overturn zone — suspended-load path.	TO BE VERIFIED: lifting procedure, crane/rigging certification, ground-bearing data and appointed-person arrangements not reviewed.	3 x 5 = 15 — HIGH	ENGINEERING: certified crane/rigging matched to a calculated lift plan (weight/CoG from vendor/drawing data — never estimated); engineered pads/mats from a ground-bearing assessment; tag lines. ADMIN: lifts categorised per the owner standard with approved plan and appointed person for complex/critical lifts; underground-service clearance for set-up areas; barricaded drop zone; trial lift and brake-hold check;	Lifting Supervisor / Appointed Person Priority: Critical Target date: [SITE INPUT REQUIRED]	1 x 5 = 5 — MODERATE	Approved lift plan incl. ground assessment; crane/rigging certificates; lifting permit Review trigger / reference: Review for every	HOLD: critical lifts over/near live plant need operations approval in addition to the lift plan; no lift without signed ground/set-up verification; settlement, rigging defect or lost communication stops the lift. [Evidence status: To be

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			load or overturn.				set-up sign-off before first lift; weather limits from the crane manual and site procedure (none assumed); no personnel under suspended loads.			complex/critical lift, after rain or relocation. Owner lifting procedure [SITE INPUT REQUIRED].	verified — no site evidence reviewed]
RA-14	Work at height, scaffold erection/alteration/ dismantling, MEWP use; dropped objects	Elevated work areas in package [CONFIRM]	Fall through open edges/penetrations; scaffold collapse from incorrect erection, ties or loading; MEWP overturn/ejection; tools/materials dropped onto personnel below.	Fatality or life-changing injury to the faller or the person struck. Persons at risk: Fitters, welders, scaffolders, inspectors at height; all personnel below.	TO BE VERIFIED: scaffold design/tag system, edge protection, anchor certification, MEWP operator certification and rescue arrangements not evidenced.	4 x 5 = 20 — EXTREME	ELIMINATION: pre-assemble at grade where practicable. ENGINEERING: collective protection first — guardrails, covered/marked penetrations, certified anchors; design check for special/suspended scaffolds; tool tethering, toe boards, barricaded drop zones. ADMIN: erection/alteration only by competent scaffolders; inspection and tag before first use, periodically and after alteration/severe weather; incomplete scaffolds physically barred; MEWP ground assessed, trained operators only, emergency lowering briefed; rescue-from-height plan and equipment at the workfront before work starts. PPE: inspected harness with double lanyard per site standard.	Scaffold Supervisor / Construction Supervisor Priority: Critical Target date: [SITE INPUT REQUIRED]	2 x 5 = 10 — HIGH	Scaffold register and tag; WAH permit (if required); rescue plan; equipment certificates Review trigger / reference: Review for every new elevated workfront; scaffolds periodically. Owner standards [SITE INPUT REQUIRED].	HOLD: no scaffold access without a current 'safe for use' tag; work at unprotected edges prohibited; any fall-arrest deployment or anchor defect stops the task. [Evidence status: To be verified — no site evidence reviewed]
RA-15	Excavation and trenching for foundations, pipelines and underground services (incl. work in/around open excavations)	Civil work areas in package [CONFIRM]	Wall collapse; strike on buried live services (cables, pipelines, fibre) not shown or wrongly marked; falls into open excavations; spoil/plant surcharge at edges.	Burial/asphyxiation fatality; electrocution or hydrocarbon release from a service strike; fractures. Persons at risk: Workers in/beside excavations, plant operators, personnel near unprotected edges.	TO BE VERIFIED: services drawings, permit practice, soil data and edge protection not reviewed.	3 x 5 = 15 — HIGH	ENGINEERING: protective system (benching/shoring/shielding) selected by a competent person for actual soil/depth — dimensions from the excavation design, not assumed; service-detection scan and trial holes before mechanical digging; rigid edge barriers, safe crossings, access ladders. ADMIN: excavation permit with services clearance signed by operations/E&I; daily and after-rain competent-person inspections; mechanical digging stops at the agreed distance from marked services, hand digging takes over; spoil/plant kept back per design; open excavations barricaded, signed and lit overnight.	Civil Supervisor / Competent Person (Excavations) Priority: Critical Target date: [SITE INPUT REQUIRED]	1 x 5 = 5 — MODERATE	Excavation permit; services-clearance drawings; scan records; inspection log Review trigger / reference: Review daily while excavations are open and after rain. Owner excavation procedure [SITE INPUT REQUIRED].	HOLD: no mechanical excavation without signed services clearance; any unmarked service found, or failed edge protection, stops and closes the excavation. [Evidence status: To be verified — no site evidence reviewed]
RA-16	Pressure testing — hydrostatic (default) and pneumatic (conditional, only where specified and justified)	Test boundaries in package [CONFIRM pressures from approved test procedures]	Failure of a test blank, hose, fitting or weak component under pressure: high-energy jet/projectiles (hydrostatic); far more violent stored-energy blast with gas (pneumatic). Test pressures are NOT stated here — they must come from the approved test pack.	Fatality/serious injury; multiple fatalities possible for pneumatic failure; equipment damage. Persons at risk: Test crew and anyone inside or beyond the test boundary.	TO BE VERIFIED: test procedures, gauge calibration records, boundary practice, and whether pneumatic testing is specified at all — not evidenced publicly.	3 x 5 = 15 — HIGH	ELIMINATION/SUBSTITUTION: hydrostatic wherever the specification allows. ENGINEERING: engineered, approved test pack (medium, pressure, hold points, vents/drains); restraints on temporary hoses/connections; calibrated gauges with current certificates. ADMIN: barricaded exclusion zone around the full boundary during pressurisation and hold (enlarged for pneumatic, sized by a specific engineered risk review with senior approval); staged pressurisation with inspection at each stage; single command for pressurisation; controlled depressurisation and drain-down. PPE: task PPE outside the line of fire only — exclusion is the control.	Mechanical / Commissioning Engineer (Test Supervisor) Priority: Critical Target date: [SITE INPUT REQUIRED]	1 x 5 = 5 — MODERATE	Approved test pack; pressure-test permit; calibration certificates Review trigger / reference: Review for every test. Approved test specification [SITE INPUT REQUIRED].	HOLD: pressurisation only with the boundary confirmed clear and recorded; pneumatic testing must not proceed without the engineered review and senior approval in the test pack. [Evidence status: To be verified — no site evidence reviewed]
RA-17	Pre-commissioning and commissioning —	NFE/NFS new trains and systems	Unexpected energisation/rotation while construction personnel still work	Fatality/serious injury from rotating equipment, pressure, electrical energy or asphyxiation.	TO BE VERIFIED: commissioning procedures, boundary control,	3 x 5 = 15 — HIGH	VERIFY FIRST: confirm purging medium/procedure; if nitrogen is used — discharge routed to a safe location, vent points barricaded, oxygen monitored at	Commissioning Lead Priority: Critical Target date:	1 x 5 = 5 — MODERATE	Commissioning permit; energisation notice;	HOLD: no energisation of any system with open construction permits on it; no entry into or breaking

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	flushing, purging, loop testing, energisation, rotating-equipment testing	under commissioning [CONFIRM boundaries]	on connected systems; inert-gas purging (if nitrogen is used — NOT confirmed) discharging into work areas creating oxygen-deficient atmospheres; flushing surges/debris.	Persons at risk: Construction crews on connected systems; commissioning technicians; personnel near purge vents and low areas.	status board and purging media not evidenced.		access points, no entry into purged systems until oxygen is proven normal by test. ENGINEERING: guards refitted before rotation tests; remote start where the design allows. ADMIN: formal boundary certificates with a single status register; energisation notices issued in advance; warning tags on energised systems; construction and commissioning permits cross-checked daily at the boundary meeting.	[SITE INPUT REQUIRED]		boundary certificate; purge plan/gas tests (if applicable) Review trigger / reference: Review daily during commissioning. Commissioning procedure [SITE INPUT REQUIRED].	into any purged system until oxygen is proven acceptable by test and recorded. [Evidence status: To be verified — no site evidence reviewed]
RA-18	First hydrocarbon introduction and start-up of new trains/systems; PSSR and handover to operations	New NFE/NFS trains and modified systems [CONFIRM]	Leaks at unproven flanges/joints, gasket failure or wrong line-up during first fill and pressurisation; debris, open punch items or blind-register errors transferred to operations at handover.	Flammable/toxic/cryogenic release, fire, fatality, damage to new and operating plant; start-up incident. Persons at risk: Commissioning team, operations staff, nearby construction crews.	TO BE VERIFIED: PSSR/handover procedure, punch-list system and leak-test regime are owner documents — not reviewed.	3 x 5 = 15 — HIGH	ENGINEERING: staged pressurisation with hold points and leak surveys at each stage; remote monitoring where practicable. ADMIN: system-by-system PSSR with operations, construction, process and HSE sign-off; no start-up with open safety-critical punch items; valve/line-up walkdown against P&IDs; blind register reconciled before repressurisation; formal boundary certificate recording when control passes from construction to operations permits; exclusion zone during first fill, non-essential personnel cleared. PPE: per commissioning procedure.	Commissioning Lead / Operations Rep. Priority: Critical Target date: [SITE INPUT REQUIRED]	2 x 5 = 10 — HIGH	PSSR certificate; leak-test records; handover / operations acceptance certificate Review trigger / reference: Review at each stage gate and handover. Owner PSSR procedure [SITE INPUT REQUIRED]; OSHA PSM [Q-10].	HOLD: no hydrocarbon introduction without PSSR closure, leak-test records and signed operations acceptance. [Evidence status: To be verified — no site evidence reviewed]
RA-19	Marine / jetty interface — LNG loading operations, jetty boil-off gas systems, work at Ras Laffan Port berths	Ras Laffan Port — six LNG berths [Q-02]; jetty boil-off recovery [Q-03][Q-05] [CONFIRM berth scope]	Release during loading-arm connection/disconnection or ship–shore interface failure; hydrocarbon vapour from boil-off systems; man overboard / fall into water during jetty work; vessel movement striking jetty equipment during maintenance.	Fire on jetty or vessel, fatalities, marine pollution; drowning. Persons at risk: Jetty operators, marine and maintenance crews at berths; vessel crews.	TO BE VERIFIED: ship–shore safety system, loading procedures, jetty permit regime and marine emergency arrangements not reviewed. Port ownership/operation by QatarEnergy is documented [Q-02].	2 x 5 = 10 — HIGH	ENGINEERING: ship–shore link and emergency shutdown/release systems function-tested before transfer; berth gas detection and firewater coverage verified. ADMIN: ship–shore safety checklist completed jointly before any transfer; berth maintenance only with berth status confirmed by Port Operations, transfers suspended where the procedure requires; work-over-water controls — lifejackets, rescue equipment and man-overboard drill for jetty tasks; weather/sea-state limits from port procedures (none assumed). PPE: lifejacket over/near water plus task PPE.	Port Operations / Marine Coordinator [titles to be confirmed] Priority: High Target date: [SITE INPUT REQUIRED]	1 x 5 = 5 — MODERATE	Port Operations authorisation; ship–shore safety checklist; owner PTW for jetty maintenance Review trigger / reference: Review for every vessel call and berth work package. QatarEnergy Ports Information [Q-02].	HOLD: no transfer with a failed ship–shore safety check; no intrusive jetty maintenance during live transfer at that berth unless Port Operations authorise it in writing. [Evidence status: To be verified — no site evidence reviewed]
RA-20	Site traffic, mobile plant and pedestrian interface; heavy-module transport movements	Plant roads, laydown areas, RLIC routes [CONFIRM traffic plan]	Vehicle–pedestrian collision, reversing in blind spots, module transport conflicting with plant traffic.	Fatality or serious injury to pedestrians, drivers and passengers. Persons at risk: Pedestrians in plant areas, drivers, passengers, other road users.	TO BE VERIFIED: traffic management plan, driver authorisation and journey management not reviewed.	4 x 5 = 20 — EXTREME	ENGINEERING: segregated walkways/crossings; one-way systems and physical separation in laydown areas; reversing alarms/cameras. ADMIN: traffic plan with speed limits set by the plan (values from the plan — not assumed); authorised-driver register; banksman for reversing in congested areas; module movements under an approved transport plan with escorts and agreed windows coordinated with operations; seat-belt rule for all occupants. PPE: high-visibility clothing in plant/traffic areas.	Construction Manager / Logistics Coordinator Priority: Critical Target date: [SITE INPUT REQUIRED]	2 x 5 = 10 — HIGH	Traffic management plan; driver authorisation register; transport plan Review trigger / reference: Review monthly and after any traffic incident/near miss. Site traffic plan [SITE INPUT REQUIRED].	STOP: any defect affecting brakes/steering/alarms removes the vehicle from service; pedestrian entry into an active movement zone stops the movement. [Evidence status: To be verified — no site evidence reviewed]
RA-	Outdoor work in	All outdoor	Heat exhaustion	Heatstroke can be fatal or	VERIFIED LEGAL	4 x 4 =	ENGINEERING: shaded/ventilated rest	HSE Manager /	2 x 4 =	Site heat-stress	STOP: outdoor work

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RA ID	Work activity / step	Asset / area / workfront	Hazard / hazardous event	Potential consequence & persons at risk	Existing controls	Initial risk (L x S = score)	Additional controls / actions (hierarchy of controls)	Responsible role & target date	Residual risk (L x S = score)	PTW / procedure reference	Verification evidence / status
21	extreme heat — heat stress, dehydration, heatstroke (Qatar summer)	workfronts	progressing to heatstroke in high temperatures; peaks in summer, during heavy work and for unacclimatised workers.	cause permanent organ damage; multiple casualties possible on peak days. Persons at risk: All outdoor workers — new arrivals, labourers, scaffolders, PPE-intensive trades especially.	CONTROL: Qatar Ministerial Decision No. (17) of 2021 prohibits work in outdoor open areas and unventilated shaded areas 10:00–15:30, annually 1 June–15 September; work must be suspended where WBGT exceeds 32.1 °C [Q-06]. Site programme itself NOT evidenced.	16 — HIGH	areas at each workfront; cold drinking water and electrolytes at the point of work. ADMIN: comply with Decision No. (17) of 2021 hours for the actual season and display the work schedule prominently; WBGT monitoring with suspension above the legal threshold [Q-06]; joint employer–worker heat-risk plans, annual heat training and medical examinations per the Ministry framework; acclimatisation for new/returning workers; buddy checks and supervisor recognition training; rehearsed heatstroke response (rapid cooling + medical evacuation). PPE: hot-weather-suitable selections where the task allows.	Contractor Supervisors Priority: Critical Target date: [SITE INPUT REQUIRED]	8 — MODERATE	procedure; WBGT log; displayed summer work schedule Review trigger / reference: Review at season start and after any heat-illness case. Decision No. (17) of 2021 [Q-06] — QATAR LAW.	stops during the legally prohibited hours in season and whenever WBGT exceeds the legal suspension threshold; any suspected heatstroke stops the crew's work and triggers medical response and programme review. [Evidence status: Legal duty verified; site programme to be verified]
RA-22	Chemical products in use (coatings, solvents, cleaning chemicals); NDT/radiography where in package — SDS/licence dependent	Coating/ maintenance locations; weld inspection locations [CONFIRM product list and ITP]	Flammable/toxic product exposure, skin/respiratory damage, solvent-vapour fire — hazards depend entirely on the actual products. Radiography (only if used): ionising radiation from an uncontrolled source or retrieval failure.	Occupational illness, fire, spill contamination; radiation overexposure of crew or bystanders. Persons at risk: Applicators and nearby workers; radiography crew and anyone entering the controlled area.	NOT CONFIRMED: no product list/SDS file and no radiography scope were available.	3 x 3 = 9 — MODERATE	VERIFY FIRST: full product list and current SDS before mobilisation; confirm from the ITP whether radiography is used and the applicable Qatar licensing route (not assumed). ENGINEERING: ventilated mixing/application areas; approved flammables storage away from ignition; radiography collimation/shielding. ADMIN: product-specific (COSHH-style) assessment per SDS, briefed in the crew's language; spill kits at use points; radiography only by licensed/authorised personnel — calculated barricaded controlled area, warning signs in workforce languages, dosimetry, survey-meter checks, source log reconciled every shift. PPE: strictly per SDS; dosimetry (monitoring, not protection).	HSE Officer / NDT Supervisor Priority: Moderate Target date: [SITE INPUT REQUIRED]	2 x 3 = 6 — MODERATE	SDS file; product-specific assessments; radiography permit and controlled-area calculation (if applicable) Review trigger / reference: Review on every new product or inspection-method change. SDS / ITP [SITE INPUT REQUIRED].	HOLD: no product may be used before its SDS assessment is complete and briefed; no radiographic exposure while anyone is inside the controlled area or the survey meter is unavailable/defective. [Evidence status: NOT CONFIRMED — verification-critical]
RA-23	Occupational-health exposures: dust/silica, noise, vibration, manual handling, slips/trips/falls; night work and fatigue	Civil, mechanical and fabrication areas; all shifts [CONFIRM exposure survey]	Respirable silica from concrete cutting/drilling; excessive noise and hand-arm vibration; musculoskeletal injury and trips over cables/hoses/debris; fatigue errors and misunderstood instructions in a multilingual workforce on night/extended shifts.	Silicosis/lung disease (chronic), hearing loss, vibration injury, strains/fractures; serious incidents in low-supervision periods. Persons at risk: Operators and nearby workers over repeated shifts; night-shift and new-starter crews.	TO BE VERIFIED: no exposure monitoring data in public sources — hygiene survey required; shift, lighting and briefing arrangements not reviewed.	4 x 3 = 12 — HIGH	ENGINEERING: wet cutting / on-tool extraction for silica tasks; low-noise/low-vibration tool selection; task lighting to the site standard at night; mechanical handling aids; cable/hose ramps and clear walkways. ADMIN: competent-hygienist exposure assessment and monitoring with health surveillance; hearing-protection zones signed from monitoring results; tool-time limits/rotation; fatigue rules from the site plan (values from the plan — not assumed); briefings in the languages crews actually understand, with comprehension checks; critical tasks (lifts, line opening, entries, first energisation) day-shift-only unless a specific night-work assessment is approved. PPE: fit-tested RPE for silica tasks; hearing protection in signed zones.	HSE Manager / Occupational Hygienist Priority: Moderate Target date: [SITE INPUT REQUIRED]	2 x 3 = 6 — MODERATE	Hygiene survey and monitoring plan; fatigue management plan; lighting survey Review trigger / reference: Review on monitoring results, task changes, and monthly for night work. Site hygiene plan [SITE INPUT REQUIRED].	STOP: dry cutting of concrete/stone is prohibited where wet/extraction controls are unavailable; blocked walkways/escape routes are cleared before work continues in that area. [Evidence status: To be verified — no site evidence reviewed]
RA-24	Emergency response interface: plant fire/gas emergency while construction and commissioning crews occupy the complex	Whole defined assessment boundary	Transient workforce not aligned with the plant emergency plan — wrong muster point, missed headcount, re-entry into a hazardous area, blocked emergency access during a plant alarm.	Multiple fatalities in a major fire/gas event; delayed rescue. Persons at risk: Every person inside the boundary, especially new and transient workers.	TO BE VERIFIED: the approved emergency plan, alarm signals, muster points and headcount system were NOT available — all are critical site inputs. RLIC city-level services publicly listed include	2 x 5 = 10 — HIGH	ADMIN: every worker inducted on the actual alarm signals, muster points and wind-direction rule from the approved plan before starting work; contractor headcount reconciled with the plant muster system (badge/gate data — Section 5); joint drills including construction crews at the frequency set by the plan; emergency access routes kept clear and audited weekly; wind socks maintained and visible at	HSE Manager / Plant Emergency Coordinator [title to be confirmed] Priority: Critical Target date: [SITE INPUT REQUIRED]	1 x 5 = 5 — MODERATE	Approved site emergency response plan; drill and induction records Review trigger / reference: Review after every drill, alarm event or	HOLD: no workfront may operate outside alarm coverage or without a briefed muster route; a failed drill triggers corrective action before the next high-risk activity. [Evidence status: To be verified — no site evidence reviewed]

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RA ID	Work activity / step	Asset / area / workfront	Hazard / hazardous event	Potential consequence & persons at risk	Existing controls	Initial risk (L x S = score)	Additional controls / actions (hierarchy of controls)	Responsible role & target date	Residual risk (L x S = score)	PTW / procedure reference	Verification evidence / status
					fire/rescue/emergency services and a medical centre [Q-01]; site arrangements still require confirmation.		workfronts. ENGINEERING: alarm coverage audible/visible at all workfronts incl. high-noise areas (beacons).			workforce change. Site emergency plan [SITE INPUT REQUIRED — critical].	
RA-25	Hydrocarbon/ chemical handling, refuelling, waste management and spill response	Refuelling points, laydown, work areas, drainage paths [CONFIRM]	Spills of fuel, condensate, chemicals or contaminated water reaching soil, drains or the marine environment; incorrect waste segregation creating fire and environmental harm.	Soil/water/marine contamination, fire, regulatory liability under Qatar environmental requirements [CONFIRM permit conditions]. Persons at risk: Refuelling and maintenance crews; the environment incl. the port/marine interface.	TO BE VERIFIED: environmental management plan, waste arrangements and drain status not reviewed.	3 x 3 = 9 — MODERATE	ENGINEERING: bunded storage sized to the site standard; refuelling on impermeable surfaces with drip trays; drain protection in chemical areas. ADMIN: spill kits at storage/refuelling points with trained responders; spill reporting and clean-up per the environmental plan; waste segregation at source, labelled containers, authorised disposal route per current Qatar requirements (confirm from the site environmental permit). PPE: per SDS.	Environmental Officer / HSE Officer Priority: Moderate Target date: [SITE INPUT REQUIRED]	2 x 3 = 6 — MODERATE	Site environmental management plan; waste transfer records Review trigger / reference: Review monthly and after any spill. Site environmental plan [SITE INPUT REQUIRED].	STOP: any spill reaching open soil, a drain or water stops the activity until contained and reported under the environmental plan. [Evidence status: To be verified — no site evidence reviewed]

NOTE ON RESIDUAL RATINGS: several major-accident rows (e.g. RA-03, RA-04, RA-06, RA-09, RA-18) retain HIGH residual scores because severity cannot be reduced — only likelihood can. Those risks are acceptable, if at all, only through the owner's formal acceptance route with all stated controls verified in place. PPE alone never justifies a rating reduction in this register.

8. Critical controls / verification checklist

Each critical barrier for a high-consequence event must be verified with evidence before the associated work proceeds. 'To be verified' means exactly that — the barrier is not credited until the evidence exists and a competent verifier has signed it.

#	Critical control / barrier	Applicability	Evidence required	Verifier (role)	Frequency	Verification status
1	Live-plant / SIMOPS interface review completed and work authorised by operations for the area and shift	To be verified	Signed SIMOPS review / permit coordination record	Area Authority	Each shift / new workfront	Not verified
2	Process isolation / positive isolation proven and LOTO applied before intrusive work	To be verified	Isolation certificate with zero-energy verification recorded	Authorised Isolator	Every intrusive task	Not verified
3	Line-opening controls: drain/vent/purge proven, temperature state confirmed, first-break technique, gas test at break point	To be verified	Line-opening permit + gas test certificate	Mechanical Supervisor	Every line break	Not verified
4	Refrigerant/LNG system data received: inventories, isolation segmentation, gas-detection coverage for systems in the package	To be verified — verification-critical (RA-04)	Process safety information pack; detection coverage plot	Process Safety Engineer	Before intrusive work on those systems	Not verified
5	Toxic-gas decision recorded from stream composition data (monitoring/respiratory protection if sour service confirmed)	To be verified — composition data required (RA-06)	Composition data sheet; signed decision note; training/monitor records if applicable	Process Safety / HSE Manager	Before intrusive work	Not verified
6	Gas testing before and during work where the owner procedure requires it (scope per process data)	To be verified	Gas test certificates; tester authorisations; monitor calibration records	Authorised Gas Tester	Per permit / continuous where required	Not verified
7	Hot-work permit, ignition control, trained fire watch and post-work fire check in place	To be verified	Hot-work permit; fire-watch log; extinguisher checks	Hot Work Supervisor	Every hot-work task	Not verified
8	Confined-space controls: classification, isolation/blinding, atmospheric testing, attendant, communications, rescue plan with equipment at the entry point	To be verified	Entry permit; test records; rescue drill/equipment checks	Entry Supervisor	Every entry	Not verified
9	Lifting controls: approved lift plan, certified crane/rigging, ground-bearing and set-up verification, barricaded drop zone, appointed person for complex/critical lifts	To be verified	Lift plan; certificates; set-up sign-off	Appointed Person (Lifting)	Every complex/critical lift	Not verified
10	Work-at-height controls: collective protection, inspected/tagged scaffolds, fall arrest with verified anchors, rescue plan at the workfront	To be verified	Scaffold tags/register; harness inspections; rescue plan	Scaffold Supervisor / HSE Officer	Daily checks; periodic scaffold inspection	Not verified

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#	Critical control / barrier	Applicability	Evidence required	Verifier (role)	Frequency	Verification status
11	Excavation controls: permit, services clearance and scan, protective system, access/egress, daily competent-person inspection	To be verified	Excavation permit; scan records; inspection log	Competent Person (Excavations)	Daily and after rain	Not verified
12	Electrical controls: authorised persons only, prove-dead recorded, temporary-power inspection and tagging	To be verified	Electrical permits; prove-dead records; inspection register	Authorised Electrical Person	Every task; periodic inspection	Not verified
13	Pressure-test controls: approved test pack, calibrated gauges, barricaded exclusion zone, single command for pressurisation, controlled depressurisation	To be verified	Test pack; calibration certificates; boundary-clear record	Test Supervisor	Every test	Not verified
14	Commissioning boundary control: boundary certificates, energisation notices, permit cross-check before energisation; purge controls if inert gas is used	To be verified	Boundary certificates; energisation log; purge records	Commissioning Lead	Daily during commissioning	Not verified
15	Ship–shore interface controls for berth work/transfers: joint safety checklist, ESD/link function test, berth status confirmed by Port Operations	To be verified — berth scope only (RA-19)	Ship–shore checklist; Port Operations authorisation	Marine Coordinator [to be confirmed]	Every vessel call / berth package	Not verified
16	Emergency readiness: alarm coverage at all workfronts, briefed muster points, headcount reconciliation, joint drills, clear emergency access	To be verified — emergency plan is a critical site input	Induction and drill records; muster reconciliation reports	HSE Manager	Drills per plan; headcount every shift	Not verified
17	Heat-stress controls: Decision No. (17) of 2021 compliance in season (hours + WBGT suspension), cold water/shade/rest, acclimatisation, heatstroke response	Applicable — legal duty verified [Q-06]; site programme to be verified	Heat-stress procedure; WBGT logs; displayed schedule; training records	HSE Manager	Daily in hot season	Not verified
18	MOC / impairment control: no bypass of safety-critical systems without approved permit and compensating measures	To be verified	Bypass/impairment register; MOC records	Process Safety Engineer	Reviewed every shift while active	Not verified
19	Contractor competency: verified qualifications/authorisations, language-appropriate induction and toolbox talks, supervision in place	To be verified	Competence matrix; induction records; authorisation registers	Construction Manager / HSE Manager	At mobilisation; periodic audit	Not verified
20	Environmental containment: bunded storage, spill kits and trained responders, drain/marine protection, waste segregation	To be verified	Inspection records; waste transfer notes; spill log	Environmental Officer	Weekly inspection	Not verified

9. Task sequencing / JSA / permit matrix

Permit names below are generic functional descriptions. The owner's current permit-to-work system governs the actual permit types, forms and approval chain — obtain the controlled site PTW procedure (Section 14) and re-map this table to it before use. A task-level JSA/JHA must be produced for each activity step by the executing contractor and briefed to the crew before work.

Task / activity	Prerequisite permit(s)	Isolation requirements	Drawings / documents	Competent persons	Pre-job briefing	Field verification	Return to service / handover
Work adjacent to live process equipment	Owner PTW; SIMOPS authorisation	As defined by operations for the task	Plot plan extract; method statement; this RA	Area Authority; Contractor Supervisor	Joint operations/contractor toolbox talk	Area Authority field check before start	Permit close-out with operations
Hot work in/near process areas	Hot-work permit; PTW	Equipment isolation as required by permit	Gas test certificate; fire-watch assignment	Hot Work Supervisor; Fire Watch; Gas Tester	Permit briefing at the work point	Gas test before and during work; fire-watch log	Post-work fire check signed; permit closed
Line opening / flange breaking (incl. cryogenic systems)	Line-opening permit; PTW	Positive isolation + drain/vent/purge proven; temperature state confirmed	P&ID; line list; isolation certificate; blind register	Authorised Isolator; Mechanical Supervisor	First-break briefing incl. line-of-fire and cold-burn controls	Two-person line ID verification; gas test at break	Reinstatement checklist; blind register reconciled
Confined-space entry	Entry permit; PTW	Blinding/isolation of all connected lines; lockout of agitators etc.	Entry plan; rescue plan; gas test records	Entry Supervisor; Attendant; Gas Tester; Rescue Team	Entrant/attendant briefing and comms check	Continuous monitoring; entry log maintained	All persons/equipment accounted for; permit closed
Complex / critical lifting (incl. expansion modules)	Lifting permit (if required by site system)	Area cleared; plant isolation if over live equipment (operations decision)	Approved lift plan; crane/rigging certificates; ground assessment	Appointed Person; Crane Operator; Riggers; Signaller	Lift briefing incl. abort signals	Trial lift; drop-zone and weather check	Lift completion report; defects logged
Excavation near services	Excavation permit	Service isolation where required by owner	Services drawings; scan records; excavation design	Competent Person (Excavations)	Crew briefing on marked services	Daily inspection; hand-dig zone enforcement	Backfill/reinstatement record
Electrical work / switching	Electrical permit / sanction for test	Isolation at all sources incl. temporary supplies; earthing per rules	Single-line diagrams; isolation certificate	Authorised Electrical Person	Briefing on boundaries and backfeed sources	Prove-dead recorded at the work point	Restoration under permit; records updated

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Task / activity	Prerequisite permit(s)	Isolation requirements	Drawings / documents	Competent persons	Pre-job briefing	Field verification	Return to service / handover
Pressure testing	Pressure-test permit	Test boundary isolation; blinds per test pack	Approved test pack; calibration certificates	Test Supervisor; Mechanical Engineer	Boundary briefing; exclusion rules	Boundary-clear confirmation before pressurisation	Depressurisation and drain record; test certificate
Commissioning / energisation / purging	Commissioning permit; energisation notice	Boundary isolation per boundary certificate	PSSR records; boundary certificates; punch list; purge plan if applicable	Commissioning Lead; Operations Representative	Boundary meeting with all affected parties	Permit cross-check before energisation	Operations acceptance certificate
Jetty / berth work and transfers	Port Operations authorisation; PTW for maintenance	Berth equipment isolation as required; transfers suspended where the procedure requires	Ship–shore safety checklist; port procedures	Marine Coordinator; Jetty Operator	Joint ship–shore briefing	ESD/link function test; checklist signed	Berth handback record with Port Operations
Work at height / scaffold use	WAH permit (if required by site system)	n/a	Scaffold tag/register; rescue plan	Scaffold Supervisor; users briefed	Toolbox talk incl. drop-zone rules	Tag verified before access; anchor checks	Scaffold handback after inspection
Radiography (only if in package)	Radiography permit	Area cleared of non-essential personnel	Controlled-area calculation; source log; licensing records	NDT Supervisor; Radiation Protection Officer [to be confirmed]	Crew and area briefing	Survey-meter verification; dosimetry issued	Source secured and log reconciled

10. Responsible-role matrix

Accountability is assigned to ROLES, never to invented individuals. Role titles below are functional descriptions; the site organisation chart and the owner PTW procedure must confirm the actual titles, appointments and authority before this document is approved.

Role	Accountability in this assessment	Authority / confirmation note
Area Authority / Operations Representative	Owns the plant area; authorises work and SIMOPS; first contact for plant upsets	Title and authority to be confirmed against the site organisation chart and PTW procedure
Permit Issuer / Permit Receiver (as defined by the site PTW system)	Issue, acceptance, suspension and close-out of permits within their competence	Exact titles and authorisation route per the controlled site PTW procedure — not assumed
Port Operations / Marine Coordinator	Berth status, ship–shore interface and port emergency interface for jetty scope	Site titles to be confirmed with the QatarEnergy port organisation
Construction Manager	Construction delivery, resources, sequencing and interface management, incl. NFE/NFS interfaces	Approves high-risk work scheduling with HSE and operations
Contractor Supervisor (each contractor)	Direct supervision of their crews; toolbox talks; first-line control verification	One named supervisor per crew per shift — names from contractor mobilisation records
HSE Manager / HSE Officer	RA ownership, inspections, incident reporting, heat-stress and hygiene programmes, drill coordination	Stop-work authority to be confirmed in the site HSE plan
Process Safety Engineer	Major-accident hazard interface, refrigerant/toxic-gas data decisions, MOC/impairment control, PSSR participation	Confirms where HAZID/HAZOP/LOPA studies are required
Mechanical Engineer / Supervisor	Piping, line opening, pressure testing and joint integrity	Approves test packs jointly with commissioning
Electrical / Instrumentation Engineer	Electrical safety, isolations, loop testing and energisation readiness	Authorised-person register maintained by this function
Lifting Supervisor / Appointed Person	Lift planning, categorisation, set-up verification and lift execution control	Appointed in writing per the owner lifting standard
Authorised Gas Tester	Atmospheric testing for permits, entries and hot work	Authorisation and calibration records current
Confined-Space Attendant / Rescue Team	Entry control, communications and rescue execution	Rescue team trained and equipped at the entry point
QA/QC Inspector	Hold-point inspection for joints, welds, coating and reinstatement	Releases hold points in the ITP only
Commissioning Lead	System completion, PSSR, energisation and handover boundaries for new trains	Single accountable authority at the construction/operations boundary
Emergency Response Coordinator [site title to be confirmed]	Alarm response, muster/headcount reconciliation, drill programme with plant operations	Interface defined in the approved site emergency plan

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11. Action tracker

No action is closed merely because it is written here. Closure requires the stated verification evidence and sign-off by the closure reviewer. Overdue Critical actions block the associated work (see hold points in Section 7).

Action ID	Linked RA	Action (specific)	Priority	Accountable role	Target	Verification evidence	Status	Closure reviewer
ACT-01	RA-01/02	Obtain approved plot plan and fix the assessment boundary, train/unit list and workfront register; re-issue this RA against them	Critical	Operations / Document Control	Before approval	Plot plan revision on file; RA Rev. 1 boundary table	Open	HSE Manager
ACT-02	RA-03/04/06	Obtain process safety information (refrigerant inventories, stream compositions incl. H2S) and record the RA-04/RA-06 control decisions for every system in the package	Critical	Process Safety Engineer	Before any intrusive work	PSI pack on file; signed decision notes	Open	HSE Manager
ACT-03	All	Confirm train-by-train current operating status from QatarEnergy (live / shutdown / repair / commissioning) and re-scope affected register rows	Critical	Operations Representative	Before approval	Written status confirmation; amended register	Open	Project Manager
ACT-04	All	Replace the provisional 5x5 matrix with the owner risk matrix and re-rate all entries in a validation workshop	Critical	HSE Manager (owner)	Before approval	Workshop record; re-rated register	Open	Project Manager
ACT-05	All	Obtain current authorized owner HSE/PTW/LOTO/hot-work/confined-space/line-opening/excavation/lifting/pressure-test/commissioning/MOC/emergency and port procedures; re-map Sections 8–9 to them	Critical	Document Control (owner)	Before approval	Controlled copies in document register	Open	HSE Manager
ACT-06	RA-24	Obtain the approved site emergency plan; extract alarm signals, muster points, routes, medical, fire and rescue provisions into the induction pack	Critical	HSE Manager	Before work starts	Emergency plan on file; induction content updated	Open	Operations Representative
ACT-07	RA-24 / Sec.5	Start daily actual-headcount capture (gate/badge + contractor returns) and reconcile with the muster system; populate Section 5.2	Critical	Contractor Admin / Security	Before work starts; then daily	Dated manpower reports; reconciliation log	Open	HSE Manager
ACT-08	RA-08/09	Verify isolation hardware, blind register practice and authorised-isolator appointments by field inspection	Critical	Area Authority	Before first line break	Inspection report; appointments in writing	Open	Mechanical Engineer
ACT-09	RA-21	Confirm the heat-stress programme: Ministerial Decision No. (17) of 2021 compliance schedule displayed, WBGT monitoring, water/shade/rest provision, acclimatisation roster	Critical	HSE Manager	Before hot season / immediately for current work	Programme document; WBGT logs started	Open	Construction Manager
ACT-10	RA-13	Complete ground-bearing assessments and crane-pad verification for all planned critical-lift set-up positions at expansion interfaces	High	Appointed Person (Lifting)	Before first critical lift	Signed set-up verifications	Open	Construction Manager
ACT-11	RA-15	Complete underground-services survey/scan for all excavation areas in the first work package	Critical	Civil Supervisor / Operations	Before excavation	Scan records; marked drawings	Open	Construction Manager
ACT-12	RA-11	Prove confined-space rescue capability (trained team + equipment) by a practical drill at the first entry location	Critical	Entry Supervisor	Before first entry	Drill report with timings and corrective actions	Open	HSE Manager
ACT-13	RA-16	Approve test packs (hydrostatic default; pneumatic only with engineered justification) and verify gauge calibration certificates	High	Mechanical / Commissioning Engineer	Before first test	Approved test packs; certificates	Open	Commissioning Lead
ACT-14	RA-19	Agree the berth-work interface protocol with Port Operations (berth status confirmation, transfer suspension rules, ship–shore checklist ownership)	High	Marine Coordinator	Before first berth work package	Signed interface protocol	Open	Operations Representative
ACT-15	RA-23	Commission an industrial-hygiene baseline survey (silica, noise, vibration, welding fume) for the package tasks	Moderate	HSE Manager	Within first month of work	Survey report; monitoring plan	Open	HSE Manager
ACT-16	RA-22	Collect the full product/SDS file for coatings, solvents and cleaning products; confirm from the ITP whether radiography is in the package	High	HSE Officer	Before product mobilisation	SDS file; assessment records	Open	Contractor Supervisor
ACT-17	RA-07	Audit the live bypass/impairment register for the interface areas; close or formally extend every overdue item	Critical	Process Safety Engineer	Before hot/intrusive work	Register audit report	Open	Operations Representative
ACT-18	All	Conduct the site walkdown and worker consultation for this assessment; record observations and amend the register	Critical	HSE Manager	Before approval	Walkdown report; consultation records; RA Rev. 1	Open	Project Manager

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12. Emergency response and rescue interface

The approved site emergency response plan was NOT available to this pre-assessment. Nothing in the table below is assumed: every operational detail is a critical site input to be extracted from that plan and briefed at induction (links to RA-24 and ACT-06). QatarEnergy's public RLIC page lists fire, rescue and emergency response services and a medical centre among the city's facilities [Q-01] — that is city-level context only and does not replace the site plan.

Emergency element	Detail required from the approved site emergency plan	Status
Emergency number / control-room contact	Exact number(s) and radio channels for raising the alarm from any workfront, incl. berths	SITE INPUT REQUIRED — critical
Alarm signals	Signal types and meanings for fire, gas release, evacuation and all-clear; audibility/visibility at high-noise workfronts	SITE INPUT REQUIRED — critical
Muster points	Locations per area (incl. jetty), wind-direction alternates, and the route to each from every workfront	SITE INPUT REQUIRED — critical
Headcount / muster reconciliation	System used (badge/gate data), who reconciles contractor numbers, and the reporting line to the incident commander	SITE INPUT REQUIRED — critical
Wind indication	Wind-sock locations and the rule for selecting upwind muster routes	SITE INPUT REQUIRED — critical
Medical provision	Clinic/medical centre arrangements [CONFIRM], ambulance provision and route to the receiving hospital [CONFIRM] — no response times are stated until evidenced	SITE INPUT REQUIRED — critical
Fire response	Plant fire team interface, hydrant/firewater status at the interfaces, extinguisher provision for construction areas	SITE INPUT REQUIRED — critical
Gas-release / refrigerant-release response	Actions on detector alarm for hydrocarbon and refrigerant releases, safe refuge locations and re-entry authority	SITE INPUT REQUIRED — critical
Marine emergency interface	Man-overboard response, jetty evacuation and port emergency coordination for berth work (RA-19)	SITE INPUT REQUIRED — critical
Rescue — confined space / work at height	Teams, equipment and drill standards (see RA-11 / RA-14)	SITE INPUT REQUIRED — critical
Communications dead zones	Radio coverage survey for remote fronts and the fallback check-in system	SITE INPUT REQUIRED — critical
Contractor–operator interface	Who commands, who accounts for contractor personnel, and how construction permits are suspended on alarm	SITE INPUT REQUIRED — critical

13. Environmental and occupational-health screening

Screening only: no exposure level is stated anywhere in this document, because no monitoring data exists in public sources. Where the screening identifies a credible exposure, an industrial-hygiene assessment/monitoring action is raised (ACT-15).

Aspect	Source activity (register link)	Screening conclusion	Control / monitoring requirement
Heat stress	All outdoor work (RA-21)	HIGH — legally regulated in Qatar [Q-06]	Decision No. (17) of 2021 compliance; WBGT monitoring and suspension; acclimatisation; hydration/shade
Cryogenic exposure	LNG/refrigerant systems (RA-05)	Credible — consequence-driven	Cryogenic PPE per site standard; warm-up proof before breaking containment
Dust / respirable silica	Concrete cutting, drilling, excavation (RA-15, RA-23)	Credible exposure — unquantified	Wet cutting/extraction; hygiene survey and monitoring; fit-tested RPE where required
Welding fume / metal fume	Welding and thermal cutting (RA-10)	Credible exposure — unquantified	LEV/extraction; consumable SDS assessment; monitoring in enclosed spaces
Noise / hand-arm vibration	Plant, tools, generators (RA-23)	Credible exposure — unquantified	Noise survey; signed hearing-protection zones; tool-time management; health surveillance
Ergonomics / manual handling	Blinds, valves, modules (RA-23)	Frequent moderate risk	Mechanical aids; task assessments; team lifts
Hydrocarbon / chemical spills	Refuelling, tie-ins, product use, jetty interface (RA-25, RA-19)	Credible — incl. marine pathway	Bunding, spill kits, drain/marine protection, reporting and clean-up per environmental plan

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Aspect	Source activity (register link)	Screening conclusion	Control / monitoring requirement
Waste generation	Construction waste, contaminated absorbents/PPE (RA-25)	Routine — controlled	Segregation at source; authorised disposal route per current Qatar requirements [CONFIRM]
Water use and sanitation (hot-climate industrial city)	Welfare provision (Sec. 5/14)	Welfare-critical in heat	Potable water at workfronts, sanitation and rest facilities verified at mobilisation [CONFIRM provision]

14. Document limitations, assumptions and site-data request list

LIMITATIONS — BLUNT STATEMENT. This document is a publicly researched pre-assessment. It is based solely on the public sources listed in Section 17 (accessed 28 October 2026) and on established HSE practice for operating LNG facilities. It has NOT been verified by a site walkdown, operator or worker interview, approved drawings, current permits, process safety information, or a shift-specific headcount. It is not an official QatarEnergy document, not QatarEnergy-approved, not a contractor method statement, and not a substitute for the controlled project HSE system, for task-level JSAs, or for specialist process studies (HAZID/HAZOP/LOPA/QRA). Controlled QatarEnergy/QatarEnergy LNG manuals were not available and were not reviewed. Any person who treats this draft as authorization to work would be misusing it.

THINNESS NOTE. Public information for this complex is strong on corporate facts (operator, train count, nameplate capacity, expansion phases, port berths, published process route) and thin on everything that makes an RA executable: no public headcount, workfront register, process conditions, refrigerant inventories, stream compositions, hazardous-area data or emergency-plan detail was found. This pack is therefore deliberately control- and verification-focused: its register rows are specific about hazards, barriers and hold points, and honest that almost every site fact remains to be supplied. That is a defensible pre-assessment, not a site-verified assessment.

ASSUMPTIONS MADE (all flagged for confirmation): (1) the selected boundary is the Ras Laffan LNG liquefaction complex including its port LNG-berth interface and the NFE/NFS expansion interfaces inside RLIC, with refinery, GTL, petrochemical and other RLIC facilities excluded as neighbours; (2) the activity list reflects the operating-plant and publicly announced expansion scope — each row applies only where the approved work pack confirms the activity; (3) the provisional 5x5 matrix in Section 6.1 is used because the owner matrix was unavailable; (4) permit and role names are generic functional descriptions pending mapping to the owner's controlled procedures; (5) train-by-train operating status follows the operator's public profile pending the Critical confirmation in ACT-03, because 2026 disruption reporting [Q-09] could not be verified against an official statement.

14.1 Site-data request list

#	Site input required	Why it matters	Owner role	Priority
1	Approved plot plan, exact assessment boundary, train/unit and workfront IDs for the LNG complex	Defines what this RA actually covers; without it every area reference stays [CONFIRM]	Operations / Document Control	Critical — before approval
2	Current work package: task list, method statements, JSA/JHA set and look-ahead schedule (incl. which NFE/NFS interfaces are live)	Confirms which register rows apply and which are N/A	Construction Manager	Critical — before approval
3	Process safety information: P&IDs, line/equipment lists, design pressures/temperatures, refrigerant types and inventories, stream compositions incl. H2S/mercaptans, SDS file	Closes the verification-critical rows RA-04/RA-06; no process values are assumed in this draft	Process Safety Engineer	Critical — before intrusive work
4	Current operating status of each LNG train (incl. any repair/restart scope reported in 2026 secondary sources)	Live, shut down, under-repair and under-commissioning trains carry different hazards; status must come from QatarEnergy, not press reports	Operations	Critical — before approval
5	Hazardous-area classification drawings and electrical single-line drawings	Govern hot-work locations, temporary power and electrical work controls	E&I Engineer	Critical — before hot/electrical work
6	Underground services drawings and survey/scan records; ground-bearing/geotechnical data	Prevent service strikes and crane set-up failures (RA-13/RA-15)	Civil / Appointed Person (Lifting)	Critical — before excavation / first critical lift
7	Owner (QatarEnergy / QatarEnergy LNG) risk matrix and risk-acceptance criteria	Replaces the provisional 5x5 matrix and bands used in this draft	HSE Manager (owner)	Critical — before approval
8	Current owner HSE manual and procedures: PTW, LOTO/isolation, hot work, confined space, line opening, excavation, lifting, pressure testing, commissioning/PSSR, MOC, emergency plan, port/jetty procedures	These controlled documents govern execution; this draft does not claim to have reviewed them	Document Control (owner)	Critical — before approval
9	SIMOPS plan and live permit register access	Needed to operate RA-02 controls and daily permit coordination	Area Authority	Critical — before work starts
10	Actual headcount by shift: gate/access-control report, contractor mobilisation register, daily manpower report	Current total persons inside the boundary is not publicly disclosed; emergency headcount depends on it	Contractor Admin / Security	Critical — before work starts; then daily

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#	Site input required	Why it matters	Owner role	Priority
11	Gas detection and fire-protection status at the interfaces; impairment/bypass register	Confirms credited barriers for hot work and line opening (RA-07)	Operations / Fire & Gas	Critical — before hot/intrusive work
12	Approved emergency response plan: alarm signals, muster points, routes, medical/ambulance provision, fire and rescue capability, communications — incl. port/marine emergency interface	RA-24 and RA-19 cannot be closed without it; no locations or response times are assumed in this draft	HSE Manager / Emergency Coordinator	Critical — before work starts
13	Lifting studies and crane/rigging certificates for the planned expansion-interface lifts	Closes RA-13 hold points for the actual lifts planned	Appointed Person (Lifting)	High — before first critical lift
14	Heat-stress programme documents: site procedure, WBGT monitoring arrangements, displayed summer schedule per Ministerial Decision No. (17) of 2021	Legal duty verified [Q-06]; site implementation evidence is still required	HSE Manager	High — before hot season / immediately for current work
15	Site walkdown report, worker consultation records, incident/near-miss history and inspection findings authorised for this assessment	Grounds the RA in observed conditions instead of public sources alone	HSE Manager	High — before approval
16	Existing specialist studies: HAZID/HAZOP reports, LOPA/QRA where they exist	An activity RA does not replace these; interfaces must be confirmed (RA-03/RA-04)	Process Safety Engineer	High

15. Approval, validation and review

15.1 Implementation, validation & review

Item	Requirement for this assessment
Critical controls / hold points	The critical controls in Section 8 and the hold point on each Section 7 register row must be verified with evidence before the associated work starts. No work within agreed protection zones of live equipment without signed operations authorisation.
Pre-start verification	PTW active, isolations proved (zero-energy verification), refrigerant/composition data received and RA-04/RA-06 decisions recorded (ACT-02), train status confirmed (ACT-03), owner matrix adopted (ACT-04), emergency arrangements briefed, toolbox talk completed and actual headcount capture running (ACT-07).
Review triggers	See Section 15.3 — any trigger event requires re-validation of the affected register rows before work continues.
Emergency and rescue arrangements	All operational details are critical site inputs from the approved site emergency plan (Section 12). Nothing in this pre-assessment states a muster point, response time or alarm level.

15.2 Consultation & approval

Prepared by / role / date	Operations / supervisor review	HSE review	Authorized approver / date
Role: HSE — competent assessor Name: _____ Signature: _____ Date: _____	Role: Area Authority / Operations Representative (Construction Manager and Port Operations review also required where applicable — see note) Name: _____ Signature: _____ Date: _____	Role: HSE Manager Name: _____ Signature: _____ Date: _____	Role: as defined by the owner's document system Name: _____ Signature: _____ Date: _____

Additional review functions to be completed in the owner's document system: Reviewed by — Construction (Construction Manager); Reviewed by — Process / Mechanical / Electrical discipline engineers (as applicable); Port Operations for jetty scope. Approval block intentionally left blank: names, signatures and dates may only be entered by the real authorized persons after site validation. Signing this Rev. 0 pre-assessment without closing the Critical site inputs in Section 14.1 would not constitute a valid approval for execution.

15.3 Review triggers

REVIEW TRIGGERS: change in scope, method, equipment or process status (including any change to a train's operating/repair/commissioning status); new SIMOPS condition; any incident or significant near miss; receipt of refrigerant inventory or composition data, or any change to an isolation boundary; new contractor or major workforce change; onset of the summer heat-regulation season or a severe-weather event; expired or failed critical control; expired permit conditions; or the review interval set by the owner's procedure — whichever occurs first.

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16. Quality check

QA check	Result
All numeric figures carry a source / date / scope label	PASS — every figure in Sections 3 and 5 is tied to a source register entry [Q-01..Q-09]
No capacity (MTPA) or third-party estimate presented as a site headcount	PASS — Section 5.1 warning; current headcount stated NOT PUBLICLY DISCLOSED; Section 5.2 left as a site-record form
No invented site count, workfront, process condition, refrigerant inventory, chemical composition, incident statistic, permit/drawing number, name, signature or approval	PASS — unknowns are [CONFIRM] / [SITE INPUT REQUIRED]; approval block blank
Boundary discipline: LNG complex only; refinery / GTL / petrochemicals / Barzan / Mesaieed / offshore not merged in	PASS — neighbours named and excluded in Sections 1.1, 3 and 4; only the documented LNG berth interface is in scope
Expansion targets labelled as targets (126 MTPA on NFE+NFS completion; 142 MTPA NFW by 2030), not current capacity	PASS — Sections 2 and 3 label both as targets; current capacity stated as 77 MTPA / 14 trains [Q-03][Q-04]
2026 disruption reporting not presented as verified fact	PASS — carried only as flagged secondary reporting [Q-09] with train status raised as Critical action ACT-03
Every High/Extreme hazard has controls, a responsible role, verification evidence and a hold point / acceptance requirement	PASS — checked row by row in Section 7 and mirrored in the XLSX register
Risk scores internally consistent with the stated matrix; no residual reduction without a credible control rationale	PASS — scores recomputed programmatically; severity is not reduced for major-accident rows
'Existing control' and 'additional action' clearly distinguished	PASS — separate columns; existing controls carry evidence status
Legal requirements vs technical guidance not misclassified; jurisdiction labelled	PASS — Qatar Ministerial Decision No. (17) of 2021 labelled QATAR LAW; ISO 45001 labelled a management-system standard; OSHA labelled US technical guidance only
Source links work; claim-level citations present	PASS for sources opened on 28 October 2026 ([Q-01]–[Q-08]); [Q-09] honestly marked partially verified and [Q-10]–[Q-13] marked as cited in the reference master prompt
Tables legible; landscape used for wide registers; headers repeat; headers/footers and page numbers present	PASS — register, checklists, permit matrix and action tracker are landscape with repeated header rows; header carries 'QATARENERGY — RAS LAFFAN LNG COMPLEX'; footer carries document ID, Rev. 0, NOT APPROVED FOR EXECUTION and page numbers
Document status states 'not approved for execution'	PASS — status banner on the cover and in every footer
Format follows the supplied generic reference system without presenting it as a client form	PASS — reference cover/doc-control grid, proposed-matrix presentation, 12-column register logic and approval block adopted; identified as a generic layout reference only, NOT an official QatarEnergy form

17. References and traceability

Access date for all sources: 28 October 2026. Verification status is stated honestly for each source. Requirement types: [LAW] Qatar legal/regulatory, [CLIENT] owner-specific, [STANDARD] international standard, [GUIDANCE] technical guidance only.

Ref	Source (title — organisation, date)	Exact URL	Assertion(s) supported in this document	Verification status
Q-01	Industrial Cities — Ras Laffan Industrial City (RLIC) — QatarEnergy, Page current at access date	https://www.qatarenergy.qa/en/Whatwedo/IndustrialCities/Pages/default.aspx	RLIC is 295 km2, about one hour by road north of Doha, and is the onshore base for processing North Field offshore gas. City facilities listed include Ras Laffan Port, common cooling water, desalinated/potable water, fire, rescue and emergency response services, electrical power, workers' accommodation and a medical centre. The LNG expansion at Ras Laffan is under execution. JURISDICTION: Qatar (site context).	Verified — opened 28 Oct 2026
Q-02	Ports Information — Ras Laffan Port — QatarEnergy, Page current at access date	https://www.qatarenergy.qa/en/MarketingAndTrading/Pages/PortsInformation.aspx	QatarEnergy is the owner, regulator and operator of Ras Laffan Port (est. 1996, 56 km2, northeastern tip of Qatar, 67 km from the North Field). Designed for export of LNG, LPG, condensates, petroleum products and sulphur; currently includes six LNG berths, six liquid product berths, one sulphur berth, six dry cargo berths, 14 support vessel berths and two offshore single point moorings; accommodates Q-Flex and Q-Max carriers. Supports the marine/jetty scope of RA-19.	Verified — opened 28 Oct 2026
Q-03	About Us — corporate structure and ventures — QatarEnergy LNG, Page current at access date	https://www.qatarenergylng.qa/english/AboutUs/Corporate-Structure/TermStoreId/1a1c8524-e9b3-43ee-b61a-d5e82f8cbf86/TermSetId/68e3750c-2365-4167-8c55-38fefe516c9a/TermId/bc012fed-6973-472c-918a-f40ce86633e6	QatarEnergy LNG (est. 1984) currently operates 14 LNG trains totalling 77 MTPA: N(1) Trains 1–3 at 3.3 MTPA each; N(2) Trains 4–5 (mega) at 7.8; N(3) Train 6 at 7.8; N(4) Train 7 at 7.8; S(1) Trains 1–2 at 3.3; S(2) Trains 3–5 at 4.7; S(3) Trains 6–7 at 8 MTPA. NFE ventures (4 new trains, 32 MTPA entitlement) and NFS ventures (2 new trains, 16 MTPA) are under construction; on completion NFE+NFS raise Qatar's liquefaction capacity from 77 to 126 MTPA (TARGET on completion — not current output). The same page lists Laffan Refinery, Ras Laffan Helium and Barzan Gas as separate ventures — treated here as neighbours/exclusions, not in-scope plant.	Verified — opened 28 Oct 2026

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Ref	Source (title — organisation, date)	Exact URL	Assertion(s) supported in this document	Verification status
Q-04	Operations — onshore facilities and the LNG process — QatarEnergy LNG, Page current at access date	https://www.qatarenergylng.qa:443/english/operations/onshore?TermStoreId=1a1c8524-e9b3-43ee-b61a-d5e82f8cbf86&TermSetId=32d4472e-903c-45bf-a326-4b9c1e544af1&TermId=2a68905c-f50f-4ebd-b24f-c4e92d4a34e1	Onshore facilities: 14 LNG trains, six mega-trains at 7.8 MTPA; Al Khaleej Gas AKG-1/AKG-2 supply approx. 2 bscfd of sales gas; products include condensate, LPG, ethane, sulphur and NGL; Helium Plants 1 and 2 total 2.1 bscf/yr. Published LNG process: condensate separation/stabilisation; staged removal of sulphur compounds, CO2 and water; propane and mixed-refrigerant chilling; main cryogenic heat exchanger to approx. -150 °C; pressure let-down to -162 °C; nitrogen removal; storage and vessel loading. This is the ONLY process basis used in this RA — it publishes no pressures, inventories or compositions, and none are assumed.	Verified — opened 28 Oct 2026
Q-05	QatarEnergy awards EPC contract for North Field South (NFS) — QatarEnergy announcement as reported — QatarEnergy announcement via World Oil, 17 May 2023	https://worldoil.com/news/2023/5/17/qatarenergy-awards-10-million-lng-expansion-contract-to-technip-energies-consolidated-contractors-company/	NFS comprises two LNG mega trains of 8 MTPA each (16 MTPA combined); EPC awarded to a Technip Energies / Consolidated Contractors Company (CCC) joint venture, value around US$10 billion; scope includes gas treatment, NGL recovery and helium extraction/refining within RLIC, carbon capture and sequestration facilities and a jetty boil-off gas recovery system. NFS with NFE increases Qatar's LNG capacity from 77 to 126 MTPA (TARGET on completion). QatarEnergy holds 75% of NFS. NOTE: the reporting page's headline '$10 million' is a typographical error; the article body states around $10 billion — the body figure is used.	Verified — opened 28 Oct 2026 (reporting of an official announcement)
Q-06	Work must be suspended at sites where WBGT index exceeds 32.1 °C: Labour Ministry (2026 Heat Stress Prevention Campaign) — The Peninsula Qatar, reporting Qatar Ministry of Labour, 2 June 2026	https://thepeninsulaqatar.com/article/02/06/2026/work-must-be-suspended-at-sites-where-wbgt-index-exceeds-321c-labour-ministry	QATAR LAW: Ministerial Decision No. (17) of 2021 regulates summer working hours in open workplaces — annually 1 June to 15 September; work is prohibited in outdoor open areas and unventilated shaded areas from 10:00 to 15:30; employers must display a clear work schedule, provide cold drinking water, shaded rest areas and suitable PPE, arrange medical examinations and trained first-aiders, measure WBGT, and suspend work where WBGT exceeds 32.1 °C. Verify the current official text with the Ministry of Labour / site legal register.	Verified — opened 28 Oct 2026
Q-07	ISO 45001:2018 — Occupational health and safety management systems — International Organization for Standardization (ISO), Published 2018-03; last reviewed and confirmed in 2024	https://www.iso.org/standard/45001	ISO 45001:2018 remains current (confirmed 2024); Amendment 1:2024 (climate action) applies; ISO lists the standard at stage 90.92 (to be revised, DIS in progress). STANDARD — management-system framework for this assessment's process; NOT a hazard checklist and NOT Qatar law.	Verified — opened 28 Oct 2026
Q-08	Minister Al-Kaabi announces raising Qatar's LNG production capacity to 142 MTPA before the end of 2030 (North Field West announcement) — QatarEnergy announcement via AAP / Cision wire, 26 February 2024	https://www.aap.com.au/aapreleases/cision20240226ae45105/	QatarEnergy announced a new LNG project from the North Field's western sector (North Field West) of about 16 MTPA, taking Qatar's total LNG production capacity to about 142 MTPA when completed before the end of the decade — a TARGET, not current capacity. Also announced additional North Field gas quantities of 240 trillion cubic feet. Used only as expansion context; North Field West onshore scope at Ras Laffan is not defined in the sources opened.	Verified — announcement text checked 28 Oct 2026 (wire reproduction of the QatarEnergy announcement)
Q-09	QatarEnergy preparing for LNG production startup, sources say (2026 shutdown/restart reporting) — Reuters (via search result), 8 April 2026	https://www.reuters.com/business/energy/qatarenergy-preparing-lng-production-startup-sources-say-2026-04-08/	SECONDARY REPORTING — NOT an official QatarEnergy statement: Reuters reported that QatarEnergy halted LNG production in March 2026 after military attacks on facilities in Ras Laffan and Mesaieed and was preparing a phased restart. Other 2026 reporting attributes statements to QatarEnergy that two LNG trains totalling 12.8 MTPA were damaged. This pre-assessment does NOT rely on these reports for any fact or control; train-by-train operating status is a CRITICAL site input (Section 14) to be confirmed from QatarEnergy directly.	Partially verified — search result reviewed 28 Oct 2026; treat as unconfirmed
Q-10	Process Safety Management — Overview — US OSHA, Current at access date	https://www.osha.gov/process-safety-management	TECHNICAL GUIDANCE ONLY (US regulator): PSM concepts used to structure major-accident interface controls (RA-01/07/18). NOT Qatar law and NOT a substitute for owner process-safety requirements.	As cited in the reference master prompt; topic page not separately opened in this check
Q-11	Process Safety Management — Hazard Recognition — US OSHA, Current at access date	https://www.osha.gov/process-safety-management/hazards	TECHNICAL GUIDANCE ONLY: hazard-recognition concepts supporting RA-03/04/06.	As cited in the reference master prompt; topic page not separately opened in this check
Q-12	Welding, Cutting and Brazing — Hazards and Solutions — US OSHA, Current at access date	https://www.osha.gov/welding-cutting-brazing/hazards-solutions/	TECHNICAL GUIDANCE ONLY: hot-work and welding-fume control concepts supporting RA-10.	As cited in the reference master prompt; topic page not separately opened in this check
Q-13	Confined Spaces — Overview — US OSHA, Current at access date	https://www.osha.gov/confined-spaces	TECHNICAL GUIDANCE ONLY: confined-space control concepts supporting RA-11.	As cited in the reference master prompt; topic page not separately opened in this check

Requirement-type key: the Qatar Ministerial Decision No. (17) of 2021 [Q-06] is QATAR LAW. QatarEnergy/QatarEnergy LNG owner procedures (not reviewed) would be CLIENT requirements. ISO 45001 [Q-07] is a STANDARD used as a management framework. OSHA material [Q-10..Q-13] is GUIDANCE from a US regulator and is never presented as Qatar law. END OF DOCUMENT — Rev. 0 — Public-source pre-assessment.

Template use note: The completed assessment must reflect the actual work scope and current site conditions. Risk ratings, legal/client references, workforce data, inspection observations and approvals must be verified by competent and authorized personnel. The generic reference format used for this document's layout does not establish compliance or client acceptance.

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